

Transition Based Closed Loop Restoration for Mobility Coupled Distribution Systems with Executable Actions

Weian Guo, Jun Min, and Wuzhao Li

Abstract—Restoration in mobility coupled distribution systems requires charging, communication, and repair decisions that remain feasible after network and crew constraints are applied. This paper develops a physically constrained coupled rollout that selects integer repair routes by their predicted service cost under a calibrated transition model. Continuous service allocations are updated from observed history and shared demand forecasts. Packet delivery, distribution system feasibility, and recorded crew completion determine when requested actions change the realized state. The analysis establishes finite horizon sensitivity for the continuous decision layer, a conditional regret bound for route selection, and constraint preservation during execution. Measured electric vehicle charging records, public outage observations, and an independent feeder support paired policy comparisons. Service cost evaluation improves on exposure based route scores. Adding threat propagation gives a smaller average improvement over a control with the same service score, with the clearest evidence under compound stress. The worst set extension does not improve the reported mean or conditional tail costs. Electrical projection reduces charging when feeder limits bind, while packet delivery and crew routing determine effective service and repair. These results connect distribution systems, communication networks, and repair operations through explicit state updates in a public data service restoration model.

Index Terms—service restoration, electric vehicle charging, distribution systems, communication networks, crew routing.

I. INTRODUCTION

Electric vehicles connect urban travel with distribution system operation.¹ Charging demand moves with daily activity, station access, road conditions, and driver choice. Its spatial pattern therefore differs from a conventional fixed load profile. Station planning and traffic power flow studies have shown that route choice and charging capacity can alter both travel and electrical states [1]–[3]. Operational pricing and navigation models extend this relation to real time charging decisions [4]–[6]. During a disruptive event, however, demand movement is only one part of the problem. Charging service also depends on feeder margin, communication availability, and the time at which a repair crew completes its work.

These dependencies give restoration a temporal and cross domain structure. Communication supports sensing, charging

coordination, and control delivery. Loss of local power may reduce that service after backup supply is exhausted. Delayed communication can then restrict charging actions and increase backlog. Road disruption can move demand toward stations whose electrical capacity is already scarce. Interdependent infrastructure studies explain why failures can spread across connected services [7], [8]. Power system resilience research further relates restoration decisions to service continuity under uncertain events [9], [10]. The resulting decision cannot be reduced to repairing the zone with the largest current loss because a moderate present threat may lie on a more consequential future path.

Demand forecasts provide useful information for this decision. Recent graph models represent spatial dependence and recurring temporal behavior in traffic series [11], [12]. Several models improve prediction under changing graph relations and limited observations [13]–[15]. Prediction error alone does not determine restoration quality. Two policies can receive the same forecast and still choose different actions because only one evaluates the propagation of road, power, and communication threats. Digital representations of physical operation provide a natural setting for this evaluation when their variables correspond to service and network constraints [16]–[18].

A further distinction concerns the quantity used to compare repair plans. Accumulated threat exposure can identify a heavily affected zone, but it does not directly specify the service recovered by completing a repair there. That service depends on the demand remaining when the crew finishes, available charging and communication resources, and backlog carried from earlier hours. Two routes can therefore have similar exposure scores but different service costs. A decision rule should compare their completion schedules through the same service equations used to value an action. Its evaluation should then separate the effect of that service objective from the additional effect of predicting propagation. This distinction also determines the mathematical object of interest. Continuous priorities admit sensitivity analysis, whereas selecting one integer route from a finite set calls for a comparison of route scores and realized costs.

The present study addresses three connected decisions. The first is the conversion of a shared demand forecast into a finite horizon restoration score that includes action dependent threat propagation. The second is the conversion of continuous service priorities into actions that satisfy packet, feeder, station, and crew constraints. The third is the separation between a

Weian Guo is with the Sino-German College of Applied Sciences, Tongji University, Shanghai, China, email guoweian@tongji.edu.cn.

Jun Min is with the College of Information Engineering, China Jiliang University, Hangzhou, Zhejiang, China, email minjun@cjlu.edu.cn.

Wuzhao Li is with Wenzhou Polytechnic, Wenzhou, Zhejiang, China, 310000, email lwzhao055@wzpt.edu.cn. Corresponding author is Wuzhao Li.

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predictive repair priority and the integer completion event that actually changes the state. Treating a requested priority as completed repair would credit service before a crew arrives. Treating an unprojected charging request as served load would create the same inconsistency on the electrical side.

This paper develops physically constrained coupled rollout, abbreviated as PC rollout. A compact graph recurrent model forecasts station arrivals and charging energy. A nonnegative transition model propagates mobility, power, and communication threats over a finite horizon. The primary central PC policy evaluates each candidate under the calibrated central transition. A separate robust extension evaluates a declared finite set of transition matrices. Requested charging and communication actions are reduced when packets miss their deadlines. Charging is then projected through an independent SMART-DS and OpenDSS calculation. Repair priorities form work orders, integer crew routes, and recorded completion events. Only executable service and completed repair enter the next realized state.

The paper makes the following theoretical and algorithmic contributions.

- It formulates an action dependent finite horizon restoration rule that selects a committed repair route by predicted service losses and backlog under a calibrated transition model. Continuous service allocations are recalculated as new observations arrive.
- It constructs an executable action algorithm that distinguishes requested service, packet delivery, distribution system projection, dispatched work orders, and integer repair completion. A capacity constrained assignment model treats station reassignment separately.
- It establishes finite horizon sensitivity for the continuous prediction and allocation layer, a conditional regret bound for a finite route portfolio, and feasibility and invariance results for executed actions.

The rest of this paper is organized as follows. Section II reviews related work. Section III presents the system model, decision rule, and mathematical results. Section IV presents the empirical study, interprets the evidence, and states data availability. Section V concludes the paper.

II. RELATED WORK

A. Electric mobility and distribution grids

Electric mobility couples travel activity with distribution system operation.¹Charging deployment models connect station access and trip demand with infrastructure placement [1], [19]. Stochastic demand models describe uncertainty and flexibility in vehicle charging [20]. Later planning studies incorporate road traffic uncertainty [21] and coordinate investments in roads, charging stations, and distribution networks [22]. Extreme fast charging models add demanding electrical constraints to station location and service design [3]. Traffic power flow formulations place route decisions and distribution constraints in a common optimization model [2], [23]. This literature establishes the physical relation between mobility and charging load.

Operational studies move the decision from capacity planning to traveler and station control. Pricing and navigation coordinate route choice with charging availability and distribution states [4]–[6]. Distributed assignment has also been studied when information and control are exchanged across cyber physical components [24]. These charging models decide capacity, prices, or routes. Restoration introduces an additional time sequence because communication delivery, feeder feasibility, and crew completion determine when a requested action becomes effective.

The timing distinction changes the mathematical role of an action. In a planning model, installed capacity is available once the investment decision is made. In a navigation model, a recommended route can enter the traffic state within the next assignment interval. A restoration request instead passes through message delivery, travel, and service before it changes local conditions. Charging coordination studies provide the demand and network relations needed for this setting [4], [24]. They leave room for an event based state update that records service only after the corresponding electrical or crew condition has been met.

B. Cyber physical resilience and digital twins

Interdependent infrastructure theory describes failure transfer across network layers and the resulting service consequences [7], [8]. Network models show that coupled systems can develop abrupt cascades [25]. Spreading process analysis supplies tools for studying the growth and control of such dynamics on graphs [26]. Power system resilience research relates these dependencies to service continuity under extreme events [9]. Proactive recovery models coordinate crew mobilization before a hurricane with equipment repair after the event [27]. Coupled power and information models also include communication traffic constraints and collaborative recovery under typhoon damage [10]. Restoration and communication dependent recovery therefore precede the present study. Its focus is their connection to charging service costs, predicted threat propagation, and recorded crew completion. Reinforcement learning offers a related approach to sequential decisions in transportation systems [28]. The present model uses this temporal view but keeps the action mechanism explicit so that a repair priority and a completed repair remain different states.

Physically constrained security research considers whether learned grid functions and adversarial inputs satisfy power balance, operating limits, and stealth requirements. Formal evaluation has been developed for intelligent security assessment and tree ensemble applications [29], [30]. Hybrid learned and classical load frequency control has also been studied under physically admissible attack models [31]. Those studies constrain cyber security analysis with grid equations. The restoration problem here instead allocates charging, communication, and repair resources over time, then subjects each request to physical and operational execution rules.

Digital representations provide a related basis for testing a decision before it is applied. General studies define the connection among a physical asset, its computational representation, and data exchange [16], [17]. Networked and industrial

models stress the need for an operational relation between the representation and the asset [18], [32]. The model in this paper represents charging service, backlog, communication capacity, threat propagation, feeder feasibility, and crew progress. These variables are used to calculate an action and its realized consequences rather than to create a visual copy of the system.

This operational use requires consistent clocks across the represented services. Demand and threat are observed at the decision interval. Packet events occur within that interval. Crew travel and repair can continue across several intervals. A continuous simulator can rank candidate actions, while the realized state must wait for the discrete event that makes the action effective. Digital representation studies support the connection between physical state and computational state [17], [18]. The present formulation uses that connection to separate the predictive state from the executed state and to keep their update rules explicit.

C. Graph forecasting for charging states

Charging forecasts estimate when and where vehicles will request service. Reviews describe the spatial and temporal structures used in traffic prediction [11], [12], [33]. Automated graph construction represents relationships that vary across time and data sets [13]. Fast spatiotemporal learning combines efficient graph operations with temporal information at several scales [34]. LocaleGN uses local traffic relationships to learn from few observations and transfer predictions across cities [14]. Self supervision and self distillation use temporal context to improve generalization [35]. Progressive graph convolution adapts spatial relations during both training and prediction [15]. Direct measurement studies nevertheless show that charging behavior retains substantial spatial and temporal variation [36].

Most forecasting models optimize prediction error, whereas restoration depends on the decision induced by a forecast. A forecast can have low average error and still assign insufficient attention to a zone on a future propagation path. The forecaster in this paper is therefore a shared state estimator. Central PC and the forecast matched baseline receive the same forecast, current threat, backlog, and realized history. Their difference is the propagation calculation used to value a candidate route. This comparison isolates the transition model from demand prediction and preserves equal information at the decision time.

Public infrastructure data also have distinct evidentiary roles. Measured charging transactions provide station demand. County outage observations provide temporal information about power events [37]. An independent distribution feeder determines whether a charging request is electrically feasible. Keeping these roles separate avoids representing records from different systems as one geographically coincident network. The method joins them through stated model variables. Charging requests and crew orders pass through packet, feeder, and completion checks. Station reassignment is evaluated separately under capacity constraints.

The literature therefore supplies four strands of the present problem. Charging studies describe demand movement and

service capacity. Infrastructure resilience studies describe propagation across connected domains. Graph forecasting estimates the near future demand state. Digital representations connect a candidate decision to operational constraints. The present algorithm combines these strands through a common information set and an event based execution rule. This combination permits a direct comparison between transition evaluation and a baseline that receives the same forecast without using the transition model.

III. SYSTEM MODEL AND RESTORATION METHOD

A. Operating state

Let $G = (V, E)$ be a charging service graph with N station zones.¹ Each station is linked to its four geographically nearest neighbours. Symmetrizing this neighbour relation, applying Gaussian distance weights, and adding self links gives W . Its row normalization is A . The realized service state during hour t is

$$x_t = (m_t, e_t, q_t), \quad (1)$$

where $m_t \in \mathbb{R}_+^N$ is mobility activity that drives charging demand, $e_t \in \mathbb{R}_+^N$ is charging energy demand, $q_t \in \mathbb{R}_+^N$ is charging backlog. Communication load and feeder load are calculated from the current service request. The separate state $z_t = (z_t^r, z_t^p, z_t^c)$ is the mobility, power, and communication threat field. Each threat field lies in $[0, 1]^N$. It is a normalized service degradation intensity, not the binary state of one breaker, line, or router. For example, $z_i^p = 0.6$ means a 60% derating in the zone level power dependent service factor. The value $z_i^r = 0.6$ produces a 60% proportional mobility impairment before the communication term. A binary outage can be represented as a high intensity limiting case. The scenario scope includes cyber induced loss of sensing and control availability and physical or service failures. The state describes service degradation without assigning a particular attack type. Before choosing an action for hour t , the controller observes backlog, threat, and charging history through hour $t - 1$. Current arrival demand is evaluated after that action is requested. The controller therefore uses the forecast state $\hat{x}_t = (\hat{m}_t, \hat{e}_t, q_t)$ and the shared forecast continuation, rather than the future realized demand in x_t .

The requested decision is

$$a_t = (u_t^{\text{ch}}, u_t^{\text{com}}, u_t^{\text{res}}). \quad (2)$$

The first term allocates charging service capacity, the second allocates communication service capacity, and the third is a continuous restoration priority used by the predictive scoring layer. It is not itself a completed repair. The feasible set is

$$\begin{aligned} \mathcal{A} = \{a_t \geq 0 \mid & \mathbf{1}^\top u_t^{\text{ch}} \leq B_{\text{ch}}, \\ & \mathbf{1}^\top u_t^{\text{com}} \leq B_{\text{com}}, \\ & \mathbf{1}^\top u_t^{\text{res}} \leq B_{\text{res}}\}. \end{aligned} \quad (3)$$

Here B_{ch} , B_{com} , and B_{res} are the available charging, communication, and normalized restoration resources during one decision interval. The continuous restoration variable is a service level priority used to order a common set of work orders. Integer crews follow explicit routes with travel times,

service times, and recorded repair completion events. At every policy hour, charging action is also projected through the SMART-DS and OpenDSS feasibility loop before service is credited and backlog is updated.

B. Grid and service equations

The continuous prediction layer uses service equations that are continuous, piecewise smooth, and Lipschitz on the stated bounded domain. The minimum, positive part, and clipping operators need not be differentiable at their switching points. For zone i and rollout step h , available communication service is

$$\bar{c}_{i,h} = u_{i,h}^{\text{com}}(1 - z_{i,h}^p)(1 - \alpha_c z_{i,h}^c), \quad (4)$$

where $\alpha_c \in [0, 1]$ measures direct communication degradation. The load placed on communication is

$$c_{i,h} = \beta_m m_{i,h} + \beta_e (e_{i,h} + \rho_q q_{i,h}) + \beta_r z_{i,h}^r + \beta_p z_{i,h}^p. \quad (5)$$

The resulting communication loss is

$$\ell_{i,h}^c = [c_{i,h} - \bar{c}_{i,h}]_+. \quad (6)$$

Communication support affects charging coordination and mobility service through

$$\eta_{i,h}^c = 1 - \min \left\{ 0.9, \frac{\ell_{i,h}^c}{1 + c_{i,h}} \right\}. \quad (7)$$

This algebraic relation is an hourly workload model. Executability is supplied by a finite buffer packet emulator with retransmission and deadline logic. If $d_{i,h} \in [0, 1]$ is the simulated fraction of control packets delivered before their deadline, the delivered charging and communication requests are

$$\tilde{u}_{i,h}^k = d_{i,h} u_{i,h}^k, \quad k \in \{\text{ch}, \text{com}\}. \quad (8)$$

The emulator keeps full service rate while backup power is available and applies power induced service derating only after backup exhaustion. Packet loss and delay therefore change the implemented action rather than appearing only as a later diagnostic. For each station, traffic multiplier, and backup duration, the main loop reads the closest simulated power and communication threat pair from a fixed response table. Squared distances use scales 0.75 and 0.375 for the two threat coordinates. This table is generated from training period activity before the validation and test experiments. It supplies packet execution fractions without fitting a queue model to test outcomes. In execution, communication support is evaluated with $\tilde{u}^{\text{com}}$ in place of u^{com} . The resulting raw charging service r_h is mapped to three phase SMART-DS loads. Bisection retains a tested feasible fraction $\alpha_h^{\text{AC}} \in [0, 1]$, giving

$$s_{i,h}^{\text{ch}} = \alpha_h^{\text{AC}} r_{i,h}. \quad (9)$$

The feeder must converge, voltages at nodes energized in the released base case must lie in $[0.95, 1.05]$ p.u., and maximum line loading must not exceed 1.0 p.u. Charging service before electrical projection is

$$r_{i,h} = \min \left\{ e_{i,h} + \rho_q q_{i,h}, \tilde{u}_{i,h}^{\text{ch}}(1 - z_{i,h}^p) \eta_{i,h}^c \right\}. \quad (10)$$

The charging backlog evolves as

$$q_{i,h+1} = [e_{i,h} + \rho_q q_{i,h} - s_{i,h}^{\text{ch}}]_+. \quad (11)$$

We impose $0 \leq \rho_q < 1$ and set $\rho_q = 0.22$. If hourly arrival energy is bounded by $\bar{e}$, then $q_{i,h+1} \leq \bar{e} + \rho_q q_{i,h}$ and consequently $q_{i,h} \leq \max\{q_{i,0}, \bar{e}/(1 - \rho_q)\}$.

Electrical outcomes are the solved node voltages and line loadings, rather than the earlier scalar feeder margin. Executable charging is determined by (9). The demand weighted mobility loss is

$$\ell_{i,h}^m = m_{i,h} (z_{i,h}^r + \beta_{mc}(1 - \eta_{i,h}^c)). \quad (12)$$

Unserved charging is defined once as $\ell_{i,h}^e = [e_{i,h} + \rho_q q_{i,h} - s_{i,h}^{\text{ch}}]_+$, and power dependent service loss is $\ell_{i,h}^p = z_{i,h}^p (e_{i,h} + \rho_q q_{i,h})$. The coefficients in these equations have fixed physical roles. The parameters β_m , β_e , β_r , and β_p convert mobility activity, charging demand, mobility threat, and power threat into communication workload. The coefficient ρ_q is backlog carryover, while $1 - z^p$ represents power availability. The coefficient β_{mc} converts communication service reduction into delayed trip equivalents. Here a trip equivalent is one charging arrival weighted by the modeled service disruption. It is not an observed travel time or an independently measured traffic delay. The stage risk is

$$L(x_h, a_h, \zeta_h) = \sum_{i=1}^N (\lambda_m \ell_{i,h}^m + \lambda_e \ell_{i,h}^e + \lambda_c \ell_{i,h}^c + \lambda_p \ell_{i,h}^p + \lambda_z [z_{i,h}^p z_{i,h}^c + z_{i,h}^r (1 - \eta_{i,h}^c)]). \quad (13)$$

The nonnegative weights λ_m , λ_e , λ_c , λ_p , and λ_z set the relative contributions of mobility delay, unmet charging demand, communication loss, power dependent service loss, and cross domain exposure. The weights carry reciprocal units for their respective losses. Thus λ_m is per trip equivalent, λ_e and λ_p are per kWh, λ_c is per communication workload unit, and λ_z is dimensionless. Their numerical values define a dimensionless comparison score, not monetary cost. Unserved charging appears only through ℓ^e . All coefficients are held fixed across policies.

For a feasible policy π , the finite horizon rollout objective at decision time t is

$$J_H^\pi(x_t, \zeta_t) = \sum_{h=0}^{H-1} \gamma^h L(x_{t+h}, a_{t+h}, \zeta_{t+h}), \quad (14)$$

$$a_{t+h} = \pi(\hat{x}_{t+h}, \zeta_{t+h}, A). \quad (15)$$

The objective in (14) sums this dimensionless stage score. The shared forecast continuation is suppressed in the policy notation. All reported experiments use $\gamma = 1$. Charging and communication allocations are recalculated hourly from the new observation and the remaining episode horizon. Crew routes are selected at the initial dispatch and remain committed throughout that episode. Thus service allocation responds to subsequent observations without assuming that a travelling crew can change its route instantaneously.

C. Demand forecasting

The forecaster estimates mobility and charging demand.

$$\hat{y}_{t,\dots,t+H-1} = f_\theta(y_{t-L}, \dots, y_{t-1}, A), \quad (16)$$

where $y_t = (m_t, e_t)$, L is the historical window, and H is the rollout horizon. The implemented model first mixes node features through A , then updates zone states using a recurrent cell, and finally maps the hidden state to the horizon. Keeping the forecaster small makes its role clear inside the larger decision model. Classical baselines and the learned model are compared in Section IV so that the decision results can be interpreted relative to the forecasting quality.

At the beginning of hour t , the last demand observation is y_{t-1} . The input series is transformed by a logarithmic map and normalized using only the training period. For a time index τ ,

$$\tilde{y}_\tau = \frac{\log(1 + y_\tau) - \mu}{\sigma}. \quad (17)$$

The graph recurrent forecaster uses one graph mixing operation before each recurrent update and layer normalization after it. With hidden state h_ℓ inside the historical window,

$$g_\ell = A\tilde{y}_{t-L+\ell-1}, \quad (18)$$

$$h_\ell = \text{LN}(\text{GRU}(W_g g_\ell + b_g, A h_{\ell-1})), \quad (19)$$

$$o = W_2 \text{GELU}(W_1 h_L + b_1) + b_2. \quad (20)$$

The output is reshaped to the forecast horizon and transformed back to demand units as $[\exp(\sigma o + \mu) - 1]_+$. Dropout is disabled during inference. The primary forecaster is trained with the Huber loss in normalized logarithmic coordinates

$$\mathcal{L}_{\text{for}} = \mathcal{L}_{\text{Huber}}(o, \tilde{y}). \quad (21)$$

Its spatial inductive bias comes from the fixed station distance graph mixing inside the recurrent model, not from an additional physical penalty. Three independent training runs are performed, and the model retained for the decision study is selected only by its validation score. Every training target lies within the training period, and every validation target lies within the validation period. No target window crosses either calendar boundary.

D. Transition model and action construction

Let $\zeta_h = (z_h^r, z_h^p, z_h^c) \in [0, 1]^{3N}$. Two transitions are kept explicit because prediction and execution have different roles. The continuous transition used only to score the requested restoration priority $u^{\text{pri}} = u^{\text{res}}$ is

$$\hat{\zeta}_{h+1} = \Pi_{\mathcal{Z}} \left(M\hat{\zeta}_h - R\psi(u_h^{\text{pri}}) + \hat{\xi}_h \right), \quad (22)$$

where $\mathcal{Z} = [0, 0.98]^{3N}$, $\Pi_{\mathcal{Z}}$ is Euclidean projection, $M \geq 0$ is the cross domain propagation matrix, and $R \geq 0$ stacks the three domain repair coefficients. The implemented saturation is $\psi_i(u) = \min\{3, [u_i/\delta]_+\}$, where δ is the per zone reference resource. Thus ψ is $1/\delta$ Lipschitz. The scorer assumes $\hat{\xi}_h = 0$ for deployable policies. Equation (22) does not update the realized state.

For execution, let $\bar{r}_{i,h} \in \{0, 1\}$ indicate that a dispatched work order in zone i has completed by the end of interval h ,

and define $D(\bar{r}_h) = I_3 \otimes \text{diag}(1 - \bar{r}_h)$. The main simulator uses

$$\zeta_{h+1} = D(\bar{r}_h) \Pi_{\mathcal{Z}} (M\zeta_h + \xi_h). \quad (23)$$

Waiting, travel, and service do not change $\bar{r}_h$. Only a recorded crew completion event changes it from zero to one. Thus no continuous priority is credited as realized repair. With $K_d = \rho_d I + \sigma A$ for $d \in \{r, p, c\}$, the implemented matrix is

$$M = \begin{bmatrix} K_r & \tau_{pr} I & \tau_{cr} I \\ 0 & K_p & \tau_{cp} A \\ \tau_{rc} A & \tau_{pc} I & K_c \end{bmatrix}. \quad (24)$$

The block structure in (24) separates persistence, spatial spread, and directed cross domain effects. The normalized adjacency A enters both within domain spread and selected cross domain paths. Boulder charging deficits estimate ρ_r and σ . EAGLE-I Boulder County outages estimate ρ_p . Aligned outage and charging records bound τ_{pr} . The packet emulator constructs the power to communication interval for τ_{pc} . Coefficients lacking joint field observations are assigned sensitivity ranges. Bootstrap and sign constrained intervals define an uncertainty set $\mathcal{U}$. The central matrix has spectral radius 1.062 and Euclidean operator norm 1.176, so no asymptotic contraction is claimed. Projection makes the implemented state set invariant and the policy is evaluated over a finite six hour horizon and a 100,000 step stress test.

PC rollout scores restoration by its action dependent marginal benefit. Let $w_{i,h}$ be a bounded exposure weight formed from the shared mobility and energy forecast, and let $\Omega = \text{diag}(\omega_r, \omega_p, \omega_c)$. For a candidate first restoration vector v , define

$$\mathcal{C}_t(v) = \sum_{h=0}^{H-1} \sum_{i=1}^N w_{i,h} \mathbf{1}^\top \Omega \hat{\zeta}_{i,t+h}(v), \quad (25)$$

where $\hat{\zeta}_{t+h}(v)$ follows (22), using v only in the first predicted transition. For reference quantum δ , the marginal benefit of repairing zone i is $b_{i,t}(M) = [\mathcal{C}_t(0, M) - \mathcal{C}_t(\delta e_i, M)]_+$. The primary policy uses the calibrated central matrix M_c , $b_{i,t}^{\text{cen}} = b_{i,t}(M_c)$. A separately evaluated robust extension uses the lower envelope

$$b_{i,t}^{\text{rob}} = \min_{M \in \mathcal{U}_V} b_{i,t}(M), \quad (26)$$

where $\mathcal{U}_V$ contains the central matrix, joint lower and upper vertices, and the endpoints of each individual coefficient (21 matrices in the tested set). The policy specific time indexed restoration score is

$$s_{i,t}^{\text{res}} = \omega_p z_{i,t}^p + \omega_c z_{i,t}^c + \omega_r z_{i,t}^r + \omega_\Gamma b_{i,t}^\pi. \quad (27)$$

Here $b^\pi = b^{\text{cen}}$ for the primary central PC method and $b^\pi = b^{\text{rob}}$ only for the robust extension. The restoration action uses a small uniform reserve and a concave risk allocation.

$$u_i^{\text{res}} = \frac{\eta B_{\text{res}}}{N} + (1 - \eta) B_{\text{res}} \frac{\sqrt{s_i^{\text{res}} + \epsilon}}{\sum_{j=1}^N \sqrt{s_j^{\text{res}} + \epsilon}}, \quad (28)$$

where $0 \leq \eta \leq 1$ is the reserve fraction. The square root reduces allocation contrasts relative to proportional weighting of the same positive scores. The reserve gives every zone at

least $\eta B_{\text{res}}/N$ without imposing an additional cap on a high priority zone. The weights ω_p , ω_c , ω_r , and ω_Γ are nonnegative constants that control direct domain exposure and the multistep action benefit. The constant ϵ is a small positive stabilizer that keeps every zone eligible for a nonzero reserve.

Charging and communication capacities are assigned with the same allocator so that all resource decisions share the same feasibility structure. For any budget $B \geq 0$, nonnegative score vector v , and reserve fraction $\eta_B \in [0, 1]$, define

$$\Phi_i(B, v, \eta_B) = \frac{\eta_B B}{N} + (1 - \eta_B) B \frac{\sqrt{v_i + \epsilon}}{\sum_{j=1}^N \sqrt{v_j + \epsilon}}. \quad (29)$$

PC rollout uses

$$u_i^{\text{ch}} = \Phi_i(B_{\text{ch}}, g^{\text{ch}}, \eta_{\text{ch}}), \quad (30)$$

$$u_i^{\text{com}} = \Phi_i(B_{\text{com}}, g^{\text{com}}, \eta_{\text{com}}), \quad (31)$$

where

$$g_{i,t}^{\text{ch}} = \sum_{h=0}^{H-1} \hat{e}_{i,t+h} + \kappa_q q_{i,t} + \kappa_{\text{ch}} (z_{i,t}^r + z_{i,t}^p + z_{i,t}^c), \quad (32)$$

$$g_{i,t}^{\text{com}} = \sum_{h=0}^{H-1} \hat{m}_{i,t+h} + \kappa_E \left(\sum_{h=0}^{H-1} \hat{e}_{i,t+h} + \kappa_q q_{i,t} \right) + \kappa_{\text{com}} (z_{i,t}^r + z_{i,t}^p + z_{i,t}^c). \quad (33)$$

Service capacity follows predicted demand, while the requested restoration priority follows the future cascade envelope. Charging and communication resources serve expected load. In the scoring layer, priority perturbs (22). In the execution layer, only the completion indicator in (23) changes realized threat. The coefficients κ_q , κ_{ch} , κ_E , and κ_{com} determine how observed backlog, current threat, and forecasted charging demand alter the time varying service scores. For the main experiment, an initial threat realization and adjacency based propagation create a policy independent set of at most 36 work orders. Four integer crews start from a common depot. Candidate routes are constructed from static, greedy, forecast matched, central PC, robust PC, and uniform priorities using the same jobs, travel matrix, service time draws, and dispatch uniforms for every policy. Forecast matched, central PC, and robust PC see exactly this same finite route portfolio. Candidate construction and final selection have distinct purposes. The exposure priorities above produce different feasible completion schedules. Selection then evaluates their predicted service cost, including the backlog left by earlier service shortfalls. For a candidate route P , let

$$\hat{Q}_t(P, M) = \sum_{h=0}^{H-1} \gamma^h L_c(\hat{x}_{t+h}^P, \hat{a}_{t+h}^P, \hat{\zeta}_{t+h}^P). \quad (34)$$

Here L_c uses the service equations with packet and electrical fractions fixed to one. The forecast issued at dispatch supplies every future demand value, and the remaining part of that forecast supplies each predicted allocation. Predicted backlog starts at the observed backlog and follows the same service recursion. Completed jobs, not continuous priorities, clear the predicted route state according to

$$\hat{\zeta}_{t+h+1}^P = D(\bar{r}_h^P) \Pi_{\mathcal{Z}}(M \hat{\zeta}_{t+h}^P). \quad (35)$$

No future innovations, later observations, sampled realized coefficients, packet outcomes, or OpenDSS results enter (34). Central PC selects the candidate with the smallest score at M_c . Robust PC instead minimizes the largest service score over the 21 matrices. The stronger forecast matched control uses the same service cost but replaces the propagation in (35) with $\hat{\zeta}_{t+h+1}^P = D(\bar{r}_h^P) \hat{\zeta}_{t+h}^P$. Thus it holds each unrepaired threat fixed while crediting the same possible completion events. The earlier current exposure and propagated exposure selectors are retained as additional controls. These comparisons separate the service objective from the effect of transition prediction. Travel uses geocoded station locations, 1.25 road circuitry, and 30 km/h. Service time scenarios are informed by observed EAGLE-I post peak recovery times. Because those county traces are not work orders, they are an explicit scenario prior rather than measured crew durations. A dispatched job changes no restoration state while waiting or being serviced. Only its logged completion event clears local threat, after which the zone remains repaired within the horizon. The mobility graph A is used for propagation. Crew travel uses the separate geocoded road distance matrix.

Algorithm 1 states the decision rule. Forecasting, propagation, service evaluation, and restoration allocation are kept as separate calculations so that the effect of each decision term can be inspected. Service allocation is updated hourly. Repair routes are committed once at the first dispatch and remain fixed during the episode.

Lemma 1. Let $N \geq 1$, $B_{\text{res}} \geq 0$, $0 \leq \eta \leq 1$, and $\epsilon > 0$. The restoration priority allocation in (28) is feasible for every nonnegative score vector.

Proof. Each score satisfies $s_i^{\text{res}} \geq 0$ and $\epsilon > 0$, so every square root in (28) is positive. Hence $u_i^{\text{res}} \geq 0$ for all zones. Summing (28) over all zones gives

$$\begin{aligned} \sum_{i=1}^N u_i^{\text{res}} &= \sum_{i=1}^N \frac{\eta B_{\text{res}}}{N} + (1 - \eta) B_{\text{res}} \frac{\sum_{i=1}^N \sqrt{s_i^{\text{res}} + \epsilon}}{\sum_{j=1}^N \sqrt{s_j^{\text{res}} + \epsilon}} \\ &= \eta B_{\text{res}} + (1 - \eta) B_{\text{res}} = B_{\text{res}}. \end{aligned} \quad (36)$$

The requested priority satisfies the nonnegativity and budget constraints. $\square$

E. Mathematical properties

The Lipschitz results below concern the continuous prediction and allocation operators before execution. They exclude integer priority rounding, packet deadline switches, OpenDSS acceptance tests and bisection, station flow active set changes, finite route argmin selection, and integer completion events. Finite route selection is treated by a score error bound, while execution constraints are treated by a separate preservation proposition. Algebraic equalities and bounds are stated in real arithmetic. Electrical feasibility means satisfaction of the stated numerical acceptance criteria.

To define the analytical map explicitly, hold the forecast continuation fixed and set the packet and electrical factors to one. Write $b = e + \rho_q q$ and let $\bar{s}$ be the resulting charging service. Then

$$F_{c,h}(x, a, \zeta) = (\hat{m}_{h+1}, \hat{e}_{h+1}, b - \bar{s}). \quad (37)$$

Algorithm 1: Physically constrained coupled rollout

Require Observed history, current backlog and threat, graph A , budgets B_{ch} , B_{com} , B_{res} , horizon H , and trained forecaster f_θ

Ensure Executable service action and event driven repair update

- 1: Predict $\hat{y}_{t,\dots,t+H-1}$ from observations through $t-1$
 - 2: Read the current mobility, power, and communication threats and backlog
 - 3: **for** each zone i **do**
 - 4: Roll out (22) at M_c with no first action and candidate δe_i
 - 5: Set $b_{i,t}^{\text{cen}} \leftarrow [C_t(0, M_c) - C_t(\delta e_i, M_c)]_+$
 - 6: Optionally evaluate $b_{i,t}^{\text{rob}}$ over $\mathcal{U}_V$ for the robust extension
 - 7: **end for**
 - 8: Allocate charging and communication service with (29)
 - 9: Compute requested restoration priority u_t^{pri} with (28)
 - 10: **if** this is the first hour of the restoration episode **then**
 - 11: Form the common work orders from the initial observed threat
 - 12: Apply packet delivery to obtain dispatched work orders
 - 13: Convert each priority by largest remainder and construct the shared route portfolio
 - 14: Evaluate each route with (34) and its completion schedule
 - 15: Select and commit the route with the smallest central model score
 - 16: **end if**
 - 17: Gate charging and communication actions by packets delivered on time
 - 18: Project charging through OpenDSS and credit only the executable service
 - 19: Propagate threat using (23), applying all recorded completions
 - 20: Observe the next state and repeat with the remaining episode horizon
-

Let L_c be (13) evaluated with this same ungated service. Neither $F_{c,h}$ nor L_c includes a variable packet fraction, an AC acceptance decision, or a crew event. The subscript h is suppressed where the same bounds apply at every step. Within this continuous analysis, x is driven by the fixed forecast, not by unrevealed future demand. Forecast histories are common to the compared trajectories. The forecaster argument also establishes continuity when those histories vary over a bounded set.

The analysis uses the Euclidean norm. The projection onto a closed convex set is nonexpansive, so

$$\|\Pi_{\mathcal{Z}}(u) - \Pi_{\mathcal{Z}}(v)\| \leq \|u - v\|. \quad (38)$$

Condition 1. On the stated bounded continuous prediction domain and for all feasible requested actions, the one step service state operator $F_c(x, a, \zeta)$ is Lipschitz in (x, ζ) with finite constants L_x and L_z .

$$\|F_c(x, a, \zeta) - F_c(\tilde{x}, a, \tilde{\zeta})\| \leq L_x \|x - \tilde{x}\| + L_z \|\zeta - \tilde{\zeta}\|. \quad (39)$$

Condition 2. The continuous stage risk L_c is Lipschitz in (x, ζ) with constants C_x and C_z for all feasible actions.

$$|L_c(x, a, \zeta) - L_c(\tilde{x}, a, \tilde{\zeta})| \leq C_x \|x - \tilde{x}\| + C_z \|\zeta - \tilde{\zeta}\|. \quad (40)$$

Lemma 2. Conditions 1 and 2 hold for the continuous service equations on the stated bounded domain. The continuous state transition and stage risk are also Lipschitz in the feasible action argument.

Proof. Let $\mathcal{D}$ denote the product of the closed state, threat, and feasible action intervals. The backlog bound in Section III gives a finite upper bound for q because $0 \leq \rho_q < 1$ and arrivals are bounded. All remaining state variables have finite upper bounds by the definition of $\mathcal{D}$. The scalar maps $r \mapsto [r]_+$ and $r \mapsto \min\{c, r\}$ are nonexpansive. Euclidean projection and componentwise clipping are also nonexpansive. Every affine map has a finite operator norm.

For bounded scalars a, a', b, b' , the product satisfies

$$|ab - a'b'| \leq |a| |b - b'| + |b'| |a - a'|. \quad (41)$$

The right hand side has finite coefficients on $\mathcal{D}$. The only quotient in the service equations has the form $g(\ell, c) = \ell/(1+c)$. If $0 \leq \ell, \ell' \leq \bar{\ell}$ and $c, c' \geq 0$, then

$$|g(\ell, c) - g(\ell', c')| \leq |\ell - \ell'| + \bar{\ell} |c - c'|. \quad (42)$$

This follows by adding and subtracting $\ell'/(1+c)$ and using $(1+c)^{-1} \leq 1$ together with $|(1+c)^{-1} - (1+c')^{-1}| \leq |c - c'|$. Explicit constants follow from these inequalities. Use common zone bounds $\bar{m}$, $\bar{e}$, and $\bar{q}$, and set $\bar{b} = \bar{e} + \rho_q \bar{q}$ and $\bar{c} = \beta_m \bar{m} + \beta_e \bar{b} + \beta_r + \beta_p$. For a zone input $(m, e, q, z^r, z^p, z^c, u^{\text{ch}}, u^{\text{com}})$, valid constants are

$$K_b = \sqrt{1 + \rho_q^2}, \quad (43)$$

$$K_c = \|(\beta_m, \beta_e, \beta_e \rho_q, \beta_r, \beta_p)\|_2, \quad (44)$$

$$K_\ell = K_c + 1 + B_{\text{com}}(1 + \alpha_c), \quad (45)$$

$$K_\eta = K_\ell + \bar{c} K_c, \quad (46)$$

$$K_s = K_b + 1 + B_{\text{ch}}(1 + K_\eta). \quad (47)$$

The product bound gives the communication capacity constant in K_ℓ . Since $0 \leq \ell^c \leq c \leq \bar{c}$, the quotient bound gives K_η . Applying the product bound to $u^{\text{ch}}(1 - z^p)\eta^c$ and then taking the minimum with b gives K_s . For the remaining cost terms, valid constants are

$$K_m = 1 + \beta_{mc} + \bar{m}(1 + \beta_{mc} K_\eta), \quad (48)$$

$$K_p = \bar{b} + K_b, \quad K_z = 3 + K_\eta, \quad (49)$$

$$K_L = \lambda_m K_m + \lambda_e (K_b + K_s) + \lambda_c K_\ell + \lambda_p K_p + \lambda_z K_z. \quad (50)$$

These bounds follow by applying the same product inequality to mobility loss, power loss, and the two cross domain products, respectively. The only variable output of (37) is $b - \bar{s}$. Summing squared zone differences therefore gives the joint constant $K_F = K_b + K_s$ for F_c . Cauchy-Schwarz over the zones gives $\sqrt{N} K_L$ for L_c . Thus the choices $L_x = L_z = L_a = K_F$ and $C_x = C_z = C_a = \sqrt{N} K_L$ are finite and satisfy Conditions 1 and 2, including their action bounds. $\square$

Lemma 3. For fixed action sequence and disturbance sequence, if two rollout trajectories start from threat fields ζ_0 and $\tilde{\zeta}_0$ and from the same physical state, then

$$\|\zeta_h - \tilde{\zeta}_h\| \leq \|M\|^h \|\zeta_0 - \tilde{\zeta}_0\|. \quad (51)$$

Proof. Let $\Delta_h = \|\zeta_h - \tilde{\zeta}_h\|$. Because the action and innovation sequences are shared, (22) and the nonexpansiveness in (38) give $\Delta_{h+1} \leq \|M\| \Delta_h$. Iteration proves (51). $\square$

Lemma 4. Under Condition 1, the service state error satisfies

$$E_h \leq \sum_{k=0}^{h-1} L_x^{h-1-k} L_z \Delta_k. \quad (52)$$

Proof. The two rollouts use the same action sequence. Condition 1 gives $E_{h+1} \leq L_x E_h + L_z \Delta_h$. Since the initial physical state is the same, $E_0 = 0$. Expanding the recursion gives $E_1 \leq L_z \Delta_0$ and $E_2 \leq L_x L_z \Delta_0 + L_z \Delta_1$. Repeating the expansion gives (52). $\square$

Theorem 1. Let $J_H(x_0, \zeta_0)$ be the discounted H step sum of L_c for a fixed feasible action sequence, and let $\tilde{J}_H(x_0, \tilde{\zeta}_0)$ be the corresponding risk for an initial threat estimate $\tilde{\zeta}_0$. For any $\alpha > 0$, define

$$\chi_\alpha = \max \{\|M\| + \alpha L_z, L_x\}. \quad (53)$$

If $\|\zeta_0 - \tilde{\zeta}_0\| \leq \varepsilon$, then

$$|J_H(x_0, \zeta_0) - \tilde{J}_H(x_0, \tilde{\zeta}_0)| \leq \varepsilon \left(C_z + \frac{C_x}{\alpha} \right) \sum_{h=0}^{H-1} (\gamma \chi_\alpha)^h. \quad (54)$$

For every finite H the sum is finite. If $\gamma \chi_\alpha < 1$, it is also uniformly bounded as the horizon increases by

$$\varepsilon \left(C_z + \frac{C_x}{\alpha} \right) \frac{1}{1 - \gamma \chi_\alpha}. \quad (55)$$

Proof. Let $\Delta_h = \|\zeta_h - \tilde{\zeta}_h\|$ and $E_h = \|x_h - \tilde{x}_h\|$. Lemma 3 and Condition 1 give the one step bounds

$$\Delta_{h+1} \leq \|M\| \Delta_h, \quad (56)$$

$$E_{h+1} \leq L_z \Delta_h + L_x E_h. \quad (57)$$

For a fixed $\alpha > 0$, define the weighted error $V_h = \Delta_h + \alpha E_h$. Multiplying (57) by α and adding (56) gives

$$\begin{aligned} V_{h+1} &\leq (\|M\| + \alpha L_z) \Delta_h + \alpha L_x E_h \\ &\leq \chi_\alpha \Delta_h + \alpha \chi_\alpha E_h = \chi_\alpha V_h. \end{aligned} \quad (58)$$

The two rollouts start from the same physical state, so $E_0 = 0$ and $V_0 = \Delta_0 \leq \varepsilon$. Induction gives $V_h \leq \chi_\alpha^h \varepsilon$ for every h . Since $\Delta_h \leq V_h$ and $E_h \leq V_h/\alpha$, Condition 2 yields

$$|L_c(x_h, a_h, \zeta_h) - L_c(\tilde{x}_h, a_h, \tilde{\zeta}_h)| \leq \left(C_z + \frac{C_x}{\alpha} \right) \chi_\alpha^h \varepsilon. \quad (59)$$

Multiplying by γ^h and summing from $h = 0$ to $H - 1$ gives (54). The infinite geometric series gives the uniform bound when $\gamma \chi_\alpha < 1$. $\square$

Condition 3. The continuous service transition and stage risk are Lipschitz in the action argument on the compact feasible set. The PC allocation map π_c before execution is Lipschitz in the forecast driven service state and threat state. Constants

are taken uniformly over the finite set of remaining episode horizons. There are finite constants L_a , C_a , L_π^x , and L_π^z such that

$$\|F_c(x, a, \zeta) - F_c(x, \tilde{a}, \zeta)\| \leq L_a \|a - \tilde{a}\|, \quad (60)$$

$$|L_c(x, a, \zeta) - L_c(x, \tilde{a}, \zeta)| \leq C_a \|a - \tilde{a}\|, \quad (61)$$

$$\|\pi_c(x, \zeta, A) - \pi_c(\tilde{x}, \tilde{\zeta}, A)\| \leq L_\pi^x \|x - \tilde{x}\| + L_\pi^z \|\zeta - \tilde{\zeta}\|. \quad (62)$$

Lemma 5. Condition 3 holds for the continuous PC map on the stated compact domain.

Proof. Lemma 2 already establishes the first two inequalities in the action argument. It remains to prove the bound for π_c . For a score vector $v \in \mathbb{R}_+^N$, set $r_i(v) = \sqrt{v_i + \epsilon}$, $S(v) = \sum_i r_i(v)$, and $p_i(v) = r_i(v)/S(v)$. The mean value theorem gives

$$\|r(v) - r(w)\|_2 \leq \frac{1}{2\sqrt{\epsilon}} \|v - w\|_2. \quad (63)$$

Moreover, $S(v) \geq N\sqrt{\epsilon}$ and $|S(v) - S(w)| \leq \sqrt{N} \|r(v) - r(w)\|_2$. Since $\|r(w)\|_2 \leq S(w)$, let $\Delta_S = |S(v) - S(w)|$. Adding and subtracting $r(w)/S(v)$ gives

$$\begin{aligned} \|p(v) - p(w)\|_2 &\leq \frac{\|r(v) - r(w)\|_2}{S(v)} + \frac{\|r(w)\|_2 \Delta_S}{S(v)S(w)} \\ &\leq \left(\frac{1}{2N\epsilon} + \frac{1}{2\sqrt{N\epsilon}} \right) \|v - w\|_2. \end{aligned} \quad (64)$$

Equation (29) is an affine function of $p(v)$. Its Lipschitz constant is therefore at most

$$(1 - \eta_B) B \left(\frac{1}{2N\epsilon} + \frac{1}{2\sqrt{N\epsilon}} \right). \quad (65)$$

The forecaster requires several additional operators. Every trained affine map has its finite matrix norm as a Lipschitz constant. The sigmoid and hyperbolic tangent have derivative bounds $1/4$ and 1 . The GRU combines these maps with gated products. Its hidden inputs remain bounded over the finite history, so the product inequality in Lemma 2 applies to every gate multiplication.

For LayerNorm with hidden width d , write $P_d = I - \mathbf{1}\mathbf{1}^\top/d$, $v = P_d h$, and $s = (\|v\|_2^2/d + \epsilon_{\text{LN}})^{1/2}$. Its normalization Jacobian is

$$\left(s^{-1} I - \frac{vv^\top}{ds^3} \right) P_d. \quad (66)$$

At $v = 0$, the bracket is $s^{-1} I$. Otherwise, it has eigenvalues s^{-1} perpendicular to v and ϵ_{LN}/s^3 along v . Both are at most $1/\sqrt{\epsilon_{\text{LN}}}$. With learned scale g and offset b_{LN} , the LayerNorm constant is at most $\|g\|_\infty/\sqrt{\epsilon_{\text{LN}}}$. Its output norm is at most $\|g\|_\infty \sqrt{d} + \|b_{\text{LN}}\|_2$, which also provides a bound for the next recurrent input.

Exact GELU is $t\Phi(t)$, where Φ and ϕ are the standard normal distribution and density. Its derivative is $\Phi(t) + t\phi(t)$, whose absolute value is at most $1 + 1/\sqrt{2\pi e}$. Dropout is the identity during inference. Input $\log(1 + y)$ has derivative at most one for $y \geq 0$, and normalization uses fixed strictly positive scales. The inverse transformation $\exp(\sigma o + \mu) - 1$ has bounded derivative on the bounded network output. Final positive clipping is nonexpansive. These facts, the hidden state bound, and induction over the fixed history prove that the

entire implemented forecaster has a finite constant on the stated domain.

The forecast exposure weights use positive fixed training means, sums, products, and clipping. They are bounded and Lipschitz by Lemma 2. For every candidate, (22) composes an affine map, the $1/\delta$ Lipschitz saturation, and a nonexpansive projection. Finite iteration and the bounded product inequality therefore prove that (25) is Lipschitz in (x, ζ) . Taking differences and positive parts preserves this property. For finite families f_j , the inequality $|\min_j f_j(v) - \min_j f_j(w)| \leq \max_j |f_j(v) - f_j(w)|$ gives the largest component constant for their minimum. The charging, communication, central PC, and robust priority scores are therefore Lipschitz. Combining their constants with (64) gives finite L_π^x and L_π^z . This proof ends at the continuous allocation. Finite route selection, packet events, the OpenDSS switch, and integer crew completion are separate operations. $\square$

Corollary 1 (continuous sensitivity before execution). Let $J_H^{\pi_c}(x_0, \zeta_0)$ and $\tilde{J}_H^{\pi_c}(x_0, \tilde{\zeta}_0)$ be the predictive risks generated by the continuous PC score and allocation layer from the same physical state and two initial threat estimates. Define

$$\bar{L}_x = L_x + L_a L_\pi^x, \quad \bar{L}_z = L_z + L_a L_\pi^z, \quad (67)$$

$$\bar{C}_x = C_x + C_a L_\pi^x, \quad \bar{C}_z = C_z + C_a L_\pi^z. \quad (68)$$

For any $\alpha > 0$, let

$$\bar{\chi}_\alpha = \max \left\{ \|M\| + \frac{\|R\| L_\pi^z}{\delta} + \alpha \bar{L}_z, \bar{L}_x + \frac{\|R\| L_\pi^x}{\alpha \delta} \right\}. \quad (69)$$

If $\|\zeta_0 - \tilde{\zeta}_0\| \leq \varepsilon$, then

$$|J_H^{\pi_c}(x_0, \zeta_0) - \tilde{J}_H^{\pi_c}(x_0, \tilde{\zeta}_0)| \leq \varepsilon \left(\bar{C}_z + \frac{\bar{C}_x}{\alpha} \right) \sum_{h=0}^{H-1} (\gamma \bar{\chi}_\alpha)^h. \quad (70)$$

Proof. Let $a_h = \pi_c(x_h, \zeta_h, A)$ and $\tilde{a}_h = \pi_c(\tilde{x}_h, \tilde{\zeta}_h, A)$. Condition 3 gives

$$\|a_h - \tilde{a}_h\| \leq L_\pi^x E_h + L_\pi^z \Delta_h.$$

Combining this bound with Conditions 1 and 3 and the restoration term in (22) yields

$$E_{h+1} \leq \bar{L}_x E_h + \bar{L}_z \Delta_h, \quad (71)$$

$$\Delta_{h+1} \leq (\|M\| + \|R\| L_\pi^z / \delta) \Delta_h + \|R\| L_\pi^x E_h / \delta. \quad (72)$$

For $V_h = \Delta_h + \alpha E_h$, the preceding two inequalities give $V_{h+1} \leq \bar{\chi}_\alpha V_h$. Since $V_0 \leq \varepsilon$, induction gives $V_h \leq \bar{\chi}_\alpha^h \varepsilon$. The stage risk difference satisfies

$$|L_c(x_h, a_h, \zeta_h) - L_c(\tilde{x}_h, \tilde{a}_h, \tilde{\zeta}_h)| \leq \bar{C}_x E_h + \bar{C}_z \Delta_h.$$

Using $E_h \leq V_h / \alpha$ and $\Delta_h \leq V_h$, then summing the discounted stage differences, gives the stated bound. $\square$

Theorem 2 (conditional finite portfolio regret). Let $\mathcal{P}$ be the nonempty finite portfolio of integer crew routes available at one decision time. Let $Q(P)$ be the realized integrated risk of route P and $\hat{Q}(P)$ its information feasible PC estimate. For $\varepsilon_Q, \delta_s \geq 0$, suppose $|Q(P) - \hat{Q}(P)| \leq \varepsilon_Q$ for every $P \in \mathcal{P}$, and the route solver returns $\hat{P}$ satisfying

$$\hat{Q}(\hat{P}) \leq \min_{P \in \mathcal{P}} \hat{Q}(P) + \delta_s. \quad (73)$$

Then

$$Q(\hat{P}) \leq \min_{P \in \mathcal{P}} Q(P) + 2\varepsilon_Q + \delta_s. \quad (74)$$

The bound in (74) separates score approximation from the finite route solver tolerance. An additive score offset does not affect route selection. Define $e(P) = Q(P) - \hat{Q}(P)$ and $\omega_Q = \max_{P \in \mathcal{P}} e(P) - \min_{P \in \mathcal{P}} e(P)$. The sharper offset independent bound is

$$Q(\hat{P}) - \min_{P \in \mathcal{P}} Q(P) \leq \omega_Q + \delta_s \leq 2\varepsilon_Q + \delta_s. \quad (75)$$

If the realized optimum P^* is unique, $|\mathcal{P}| > 1$, and

$$\Delta_Q := \min_{P \in \mathcal{P} \setminus \{P^*\}} \{Q(P) - Q(P^*)\} > 2\varepsilon_Q + \delta_s, \quad (76)$$

then $\hat{P} = P^*$.

Proof. Let $P^* = \arg \min_P Q(P)$. Uniform approximation and solver tolerance give $Q(\hat{P}) \leq \hat{Q}(\hat{P}) + \varepsilon_Q \leq \hat{Q}(P^*) + \delta_s + \varepsilon_Q \leq Q(P^*) + 2\varepsilon_Q + \delta_s$. Subtracting $Q(P^*)$ in the identity $Q = \hat{Q} + e$ instead gives

$$\begin{aligned} Q(\hat{P}) - Q(P^*) &= \hat{Q}(\hat{P}) - \hat{Q}(P^*) + e(\hat{P}) - e(P^*) \\ &\leq \delta_s + \omega_Q. \end{aligned} \quad (77)$$

Since $|e(P)| \leq \varepsilon_Q$, its range is at most $2\varepsilon_Q$, which proves (75). For any $P \neq P^*$, the gap condition implies $\hat{Q}(P) > \hat{Q}(P^*) + \delta_s$, so no such plan can be returned. $\square$

Proposition 1 (execution feasibility and invariance). Assume the base feeder operating point is feasible, packet enactment satisfies $0 \leq d_{i,h} \leq 1$, service arrivals are bounded, $0 \leq \rho_q < 1$, and every accepted work order has a finite integer route with finite nonnegative travel and service times. Assume also $\zeta_0 \in \mathcal{Z}$, $q_0 \geq 0$, and the allocator conditions of Lemma 1. Then each step satisfies the three resource budgets, packet reduction preserves those budgets, the OpenDSS projection returns an accepted charging action, backlog remains bounded, and $\zeta_h \in \mathcal{Z}$ for all h . In the separate station reassignment problem, nonnegative receiving capacities and an unserved demand variable ensure a feasible capacity respecting flow.

Proof. Lemma 1 applies to each of the three allocators. Hence the requested actions are nonnegative and their component sums equal the corresponding budgets. Integer conversion preserves the restoration total. For a positive integer K of crews and normalized priorities p_i , let $k_i = \lfloor K p_i \rfloor$. The integer remainder $r = K - \sum_i k_i$ lies between zero and $N - 1$. Adding one to the r largest fractional parts gives nonnegative integers summing to K . Rescaling them by B_{res}/K restores the requested resource total. For any component $u_i \geq 0$ and $d_i \in [0, 1]$, one has $0 \leq d_i u_i \leq u_i$. Summing this inequality proves that packet reduction cannot increase a budget. The feeder calculation selects a common scale $\alpha^{\text{AC}} \in [0, 1]$. The point $\alpha^{\text{AC}} = 0$ is the feasible base operating point. Bisection stores the feasible endpoint of every interval, so its returned endpoint is feasible and cannot increase the charging budget. The accepted endpoint is reapplied and solved again before return. If this repeat fails, the implementation reduces the endpoint, with the feasible zero charging point as its final fallback. Every returned action therefore passes the acceptance test. No monotonicity of AC feasibility or maximality of the

retained fraction is required for this conclusion. In the separate station flow problem, setting every assigned flow to zero and every unserved variable to its corresponding demand gives a feasible point. Any accepted solution respects the receiving capacity inequalities by construction.

Let $e_{i,h} \leq \bar{e}$. From the backlog recursion,

$$q_{i,h+1} \leq \bar{e} + \rho_q q_{i,h}, \quad (78)$$

$$q_{i,h} \leq \rho_q^h q_{i,0} + \bar{e} \frac{1 - \rho_q^h}{1 - \rho_q} \quad (79)$$

$$\leq \max \left\{ q_{i,0}, \frac{\bar{e}}{1 - \rho_q} \right\}. \quad (80)$$

The second line follows by induction, and the final line follows because it is a convex combination of $q_{i,0}$ and $\bar{e}/(1 - \rho_q)$. Thus backlog remains bounded. Suppose $\zeta_h \in \mathcal{Z}$. Projection maps $M\zeta_h + \xi_h$ into $\mathcal{Z}$. Every diagonal entry of $D(\bar{r}_h)$ is zero or one, so multiplication by $D(\bar{r}_h)$ also maps $\mathcal{Z}$ into $\mathcal{Z}$. Since $\zeta_0 \in \mathcal{Z}$, induction proves state invariance. Finally, a finite integer route assigns a finite arrival and service order to each accepted work order. The completion indicator changes only at the corresponding recorded completion time, which makes every realized repair update well defined. $\square$

Theorem 1 and Corollary 1 concern finite horizon sensitivity in the continuous prediction and allocation layer. Theorem 2 gives a conditional bound for the finite route portfolio in terms of ε_Q and the solver tolerance. Proposition 1 covers constraint preservation in execution. The numerical checks in Section IV evaluate these same properties for the reported scenarios.

IV. EXPERIMENTAL STUDY

A. Evidence sources and preprocessing

The evaluation separates measured demand, outage context, electrical feasibility, communication enactment, and crew execution instead of deriving all layers from one mobility proxy.¹ Figure 1 shows the resulting closed loop, and Table I records the role of each source.

The primary demand source is the City of Boulder Electric Vehicle Charging Station Data under a CC0 license. Its metadata defines one row as one transaction and provides local transaction time, station identity, charging duration, and delivered energy. The local archive contains all 148,136 transactions released from 2018 through 2023, with 50 station records and 1,252,719.654 measured positive kWh. Four records lack an end time, 16,113 report zero energy, no duplicate source identifier or full row was found, and no negative energy was retained. Arrival counts are exact transaction counts by start hour. Session energy is measured. Its hourly load profile is obtained by spreading each session's energy uniformly over the reported charging duration. Conservation checks recover all 148,136 arrivals and the measured positive energy to within 0.001 kWh.

Time was split before model fitting. The period from 2018 through 2021 was used for training, 2022 was used for model selection, and 2023 was used for final testing. No random row split crosses this calendar boundary. A second five fold spatial test withheld contiguous west to east station blocks from target fitting. In that inductive deployment test, a held out station's

calibration before 2022 and observed 1, 24, and 168 h lags were permitted, but its target values never entered coefficient fitting.

Power event evidence came from the CC BY 4.0 EAGLE-I release [37]. The source record contained 17 files and 249,316,543 rows. Preprocessing retained 819,528 observations for Boulder County and its six adjacent counties, including 171,187 Boulder observations. EAGLE-I timestamps are UTC and were converted to America/Denver only for alignment with local charging records. Because the source omits zero outage records and cannot distinguish a true zero from a collection gap, absent county and time pairs were preserved as missing rather than silently imputed as zero. Events were evaluated at 50, 100, and 500 customers out. At the 50-customer threshold, 2,535 events had median duration 1.75 h and 90th percentile duration 6.5 h. The post peak half recovery time was observable for 19.6% of events and had median 0.75 h. These traces inform a scenario distribution, not station level failure labels or utility work orders.

The independent electrical test uses one complete SMART-DS v1.0 San Francisco PIU feeder and its unmodified OpenDSS model. The solved circuit contains 80 buses, 219 nodes, 27 loads, 67 lines, and 13 transformers. Eighteen nodes already deenergized in the released base case are identified and excluded from active voltage minima. They are not counted as failures caused by EV action. The base active node voltage minimum is 0.989 p.u. and maximum line loading is 0.912 p.u. Requested EV actions are aggregated as balanced three phase, wye connected constant PQ increments at the 17 eligible three phase loads. Active power is the executed station request and reactive power follows a 0.98 lagging power factor. The released load voltage level (0.48 or 12.47 kV) and base load parameters are otherwise unchanged. Twenty station to load mappings are prepared in advance. Mapping $s \bmod 20$ is shared by all policies in scenario s and recorded in the scenario table. At every simulated hour, station level executable charging is aggregated at the assigned loads, OpenDSS solves the unprojected action, and a bisection backtracks its common fraction until voltage lies in $[0.95, 1.05]$ p.u., line loading is at most 1.0 p.u., and the circuit converges. Only this projected energy enters served energy and backlog equations. This gate validates the electrical feasibility of a proposed charging action. It does not simulate feeder topology reconfiguration, switching or breaker operations, fault isolation, feeder resupply, or equipment parameter changes after repair. The experiment therefore studies charging service restoration under a fixed feeder topology.

B. Demand forecasting and spatial transfer

Table II reports the fixed 2023 temporal test. The graph recurrent forecaster achieved arrival MAE 0.110 and energy MAE 0.948 kWh, compared with 0.137 and 1.101 kWh for the historical mean. Its energy RMSE (3.119 kWh) did not beat the historical mean (3.026 kWh), so no uniform metric dominance is claimed. Across three training runs, arrival MAE ranged from 0.1097 to 0.1101 and energy MAE from 0.9449 to 0.9492 kWh. The reported model was selected by its validation score, not by the smallest test error among these runs.

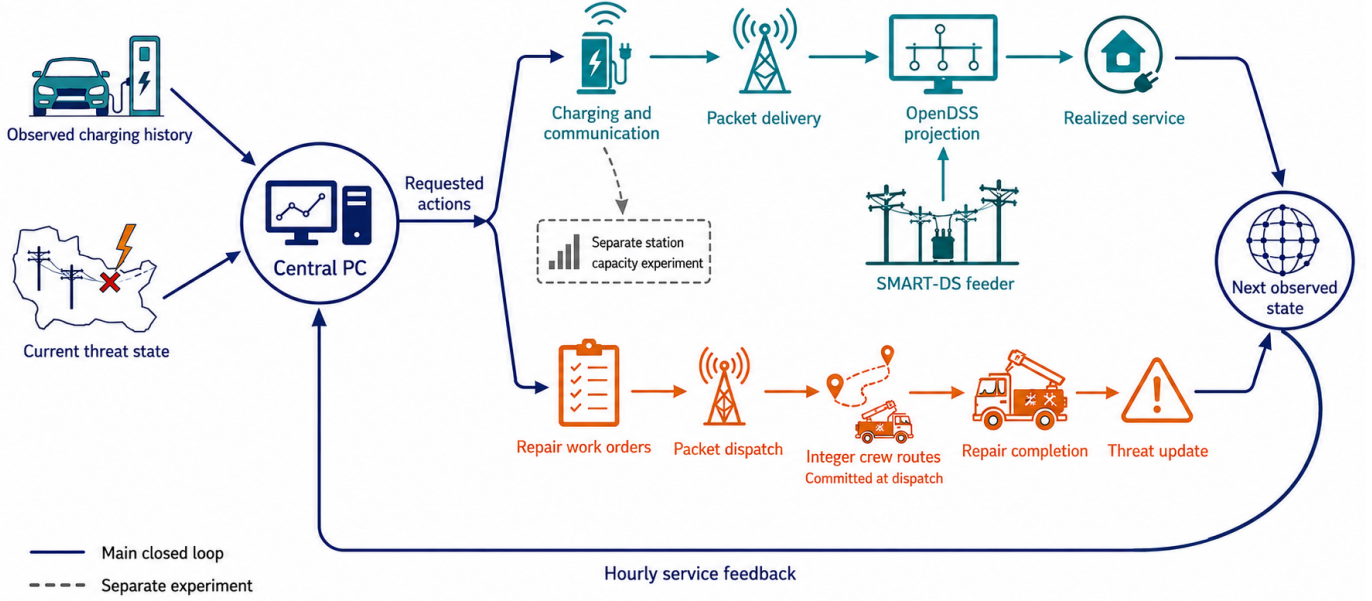


Fig. 1. Executable PC closed loop. Observed charging history supplies the shared forecast, and the controller receives the current scenario threat state. EAGLE-I observations inform transition calibration and the repair duration prior. Requested service actions pass through packet delivery and SMART-DS/OpenDSS projection. Repair work orders pass through packet dispatch, integer routing, and recorded completion events. Routes are committed at dispatch, while service allocations are updated hourly. Realized service and threat updates form the next observed state. The dashed path denotes the separate station capacity experiment and is not part of the primary 4,096-scenario loop. The conceptual illustration was generated and edited with OpenAI image tools, <https://openai.com/>.

TABLE I
INDEPENDENT EVIDENCE SOURCES AND THEIR DISTINCT ROLES

Source	Scale	Observed fields	Role in this study
City of Boulder EV sessions	148,136 transactions from 50 stations between 2018 and 2023	Arrival time, charging duration, energy (kWh), station and address	Measured charging demand with calendar periods for training, selection, and testing
EAGLE-I	249,316,543 national rows with 819,528 retained in Boulder and six adjacent counties between 2014 and 2025	County, UTC timestamp, customers out	Power state persistence, event and recovery distributions, and uncertainty bounds with missing rows preserved
SMART-DS SFO PIU feeder	80 buses, 219 nodes, 27 loads, 67 lines, 13 transformers	OpenDSS circuit, equipment and peak load operating point	Online projection for all primary policy hours and the integrated electrical stress test
Packet and route models	12,000 packet stress rows and 4,096 paired main scenarios with up to 36 jobs and four crews	Queue events, retries, deadlines, backup, integer crews, travel, service and completion times	Requested controls and threat updates after a recorded repair completion event

TABLE II
CALENDAR SPLIT TEST PERFORMANCE ON MEASURED BOULDER CHARGING SESSIONS IN 2023

Method	Arrivals		Energy (kWh)	
	MAE	RMSE	MAE	RMSE
Historical mean	0.137	0.517	1.101	3.026
Weekly seasonal	0.156	0.637	1.130	3.299
Persistence	0.175	0.678	1.059	3.186
Graph recurrent forecaster	0.110	0.527	0.948	3.119

The spatial test pooled all five held out blocks and six horizons. The spatial ridge model reduced energy MAE from 1.130 to 1.054 kWh and RMSE from 2.928 to 2.272 kWh relative to

station specific weekly history. Arrival RMSE also decreased from 0.557 to 0.440, but arrival MAE (0.161) was worse than the pooled calendar mean (0.139). Thus the measured data forecast supports temporal deployment and improves several unseen station endpoints. Arrival MAE remains higher at sparse stations, which identifies the forecast condition for interpreting the decision study.

C. Propagation calibration and policy comparison

The coefficient ledger in Table III distinguishes estimated, observational, constructed, and sensitivity range quantities. A nonnegative station deficit regression estimated mobility persistence and spatial spread from 1,268,661 training samples.

TABLE III
PROPAGATION COEFFICIENTS AND PRESPECIFIED UNCERTAINTY SET

Parameter	Central	Interval	Evidence class
ρ_r	0.728	[0.546, 0.911]	estimated
σ	0.233	[0.116, 0.349]	estimated
ρ_p	0.772	[0.754, 0.787]	estimated
τ_{pr}	0.000	[0.000, 0.032]	estimated observational
τ_{pc}	0.427	[0.169, 0.551]	constructed
ρ_c	0.500	[0.300, 0.800]	sensitivity only
τ_{rc}	0.040	[0.000, 0.080]	sensitivity only
τ_{cr}	0.080	[0.000, 0.160]	sensitivity only
τ_{cp}	0.040	[0.000, 0.080]	sensitivity only

A separate county level regression checks the aligned outage and charging series. On 3,108 aligned 2023 test hours, that regression’s road state RMSE was 0.306 versus 0.374 for persistence. It is not the station propagation model used to construct M . EAGLE-I estimated power persistence at 0.772 with a 90% day block bootstrap interval [0.754, 0.787]. The aligned coefficient from power state to EV deficit was effectively zero with upper interval 0.032, providing no evidence for a strong direct conditional association. The packet model constructed the power to communication interval from 12,000 training period packet scenarios. Its fitted central coefficient is 0.427 and its hour block bootstrap interval is [0.169, 0.551]. Directions without joint field traces span sign constrained sensitivity ranges including zero.

Central PC scores each route using the continuous service objective in (34) under the calibrated transition. Forecast matched uses the same objective, demand forecast, backlog update, completion schedule, and route candidates, but holds unrepaired threats at their current values. Robust PC instead minimizes the largest predicted service cost over 21 declared matrices. The two exposure score controls retain the earlier route ranking rules, one based on current exposure and one on propagated exposure. This comparison separates the contribution of service cost scoring from the additional effect of propagation. All five selectors share the same candidate set, including candidates constructed from central and worst set priorities. Only their final scoring rule changes. Static and greedy use their own direct route priorities, and the oracle alone receives future demand and innovations. Table IV gives the eight policy definitions and their common execution conditions.

The service score was compared with exposure scoring on 512 validation scenarios from 2022 before the revised test. Central service scoring reduced cost by 8.707 relative to exposure matched, with a paired interval [−11.113, −6.466]. Its difference from the same service score without propagation was −0.095 with interval [−0.950, 0.754]. The score definition, weights, budgets, and forecast selection rule were then fixed for the test, which retained both exposure controls and the stronger service matched control. Normalization and budgets use only 2018 through 2021 station statistics. Forecast selection uses 2022 error, and every target window lies entirely within its declared calendar period. Earlier revision checks had examined results from the same test year. The present comparisons evaluate the subsequently fixed specification on

that year.

The test comprised 4,096 paired scenarios with 1,024 per threat class. Realized coefficients were drawn independently from their stated ranges for each scenario and were shared across policies. The primary comparison family contains central PC versus each of the three service and exposure controls, both overall and in the four classes. We report paired bootstrap intervals with 20,000 resamples, paired t tests, and two sided Wilcoxon tests. Holm correction includes all fifteen comparisons separately for each test. Bootstrap intervals are marginal intervals, not simultaneous intervals for the whole family. Table XV retains every comparison. The appendix reports each value, unit, role, calibration rule, selection source, and tested range in Tables XII, XIII, and XIV.

Table V reports the primary paired comparison and the policy summary. Central PC reduced mean cost by 0.1027% relative to the same service score without propagation. The paired difference was −1.1436 with 95% interval [−1.5257, −0.7704], and the Holm adjusted paired t and Wilcoxon probabilities were 1.38×10^{-8} and 9.79×10^{-23} . The OOD compound class showed a 0.1979% reduction with difference interval [−4.8554, −2.5890]. The nominal, single domain, and cascade mean intervals included zero. Their reductions were −0.0002%, 0.0132%, and 0.0741%, respectively. Thus the overall effect is statistically distinguishable but small, and a uniform classwise improvement is not established. These intervals quantify Monte Carlo scenario uncertainty conditional on the public data and simulation model rather than variation across utilities.

The larger effect came from evaluating service rather than exposure alone. Central PC reduced mean cost by 0.6872% relative to exposure matched and 1.0753% relative to exposure central, with difference intervals [−8.3843, −7.0077] and [−13.1616, −11.0704]. Both overall comparisons remained significant after Holm correction. The service matched control also had lower mean cost than either exposure control. These results locate most of the gain in the route objective, with a smaller additional contribution from propagation. The oracle mean was 1053.67 compared with 1112.54 for central PC, leaving a 5.59% policy to oracle gap. This is an information reference, not a bound on the globally optimal restoration schedule.

Robust PC had mean cost 1112.90. Its difference from central PC was 0.3670, with marginal interval [0.0788, 0.6627]. Its upper 5% and 1% conditional mean costs exceeded central PC by 2.85 and 4.73, respectively, while the maximum observed cost was identical at 4494.26. The conditional means include a fractional observation at the empirical quantile boundary. These descriptive tail comparisons do not establish a tail protection benefit, and the worst set calculation remains a sensitivity extension rather than the principal method. Table XVII reports the empirical cost tails of all eight policies.

Table VI changes one objective weight at a time and reruns candidate selection and execution on the same 1,024 scenario inputs. The reference row is the corresponding subset of the primary experiment. Across the five changed objectives, central PC reductions ranged from 0.0597% to 0.1264%, with marginal mean difference intervals below zero. The direction

TABLE IV

POLICY INFORMATION AND EXECUTION CONDITIONS. “CURRENT” MEANS THE OBSERVATION AVAILABLE AT THE DECISION HOUR. THE COMMON ROUTE PORTFOLIO CONTAINS ROUTES INDUCED BY STATIC, GREEDY, FORECAST MATCHED, CENTRAL PC, ROBUST PC, AND UNIFORM PRIORITIES. FUTURE INNOVATIONS ARE NEVER REVEALED TO DEPLOYABLE POLICIES.

Policy	Demand information	Threat and backlog	Multistep transition rollout	Transition matrix used	Future innovations	Repair route rule	Common gates	execution
Static	Historical station mean	Backlog only for charging	No	None	No	Direct route from static priority	Yes	
Greedy	Common forecast	Current threat and backlog	No	None	No	Direct route from current threat priority	Yes	
Forecast matched	Common forecast	Current threat and backlog	No	None in score	No	Service cost with unrepaired threat held fixed	Yes	
Central PC	Common forecast	Current threat and backlog	Yes	Calibrated central M	No	Predicted service cost over common portfolio	Yes	
Robust PC (extension)	Common forecast	Current threat and backlog	Yes	Worst score over 21 declared matrices	No	Worst predicted service cost over common portfolio	Yes	
Exposure matched	Common forecast	Current threat and backlog	No	None in score	No	Current exposure weighted completion over common portfolio	Yes	
Exposure central	Common forecast	Current threat and backlog	Yes	Calibrated central M	No	Accumulated threat exposure over common portfolio	Yes	
Oracle	Realized demand	Current threat and backlog	Yes	Central M	Yes	Direct route from oracle priority	Yes	

TABLE V

CENTRAL PC COMPARED WITH THE SAME SERVICE SCORE WITHOUT THREAT PROPAGATION, FOLLOWED BY ALL POLICY MEANS. POSITIVE REDUCTION DENOTES LOWER COST THAN FORECAST MATCHED. Δ IS CENTRAL MINUS MATCHED COST. INTERVALS ARE PAIRED MARGINAL INTERVALS. ALL POLICIES SHARE 4,096 SCENARIOS AND THE ORACLE ALONE OBSERVES FUTURE DEMAND AND INNOVATIONS.

Group	Pairs	Reduction (%)	Δ	Δ 95% CI
All	4096	0.1027	-1.1436	[-1.5257, -0.7704]
Nominal	1024	-0.0002	0.0013	[-0.0042, 0.0075]
Single domain	1024	0.0132	-0.1021	[-0.2602, 0.0510]
Cascade	1024	0.0741	-0.7946	[-1.7624, 0.1327]
OOD compound	1024	0.1979	-3.6789	[-4.8554, -2.5890]

Policy	Mean cost	Reduction (%)
Forecast matched	1113.68	0.0000
Central PC	1112.54	0.1027
Robust PC	1112.90	0.0697
Exposure matched	1120.23	-0.5885
Exposure central	1124.63	-0.9832
Static	1202.74	-7.9970
Greedy	1122.54	-0.7960
Oracle	1053.67	5.3887

of the comparison therefore persisted under these weight changes. However, none of their paired mean tests remained significant after correction across the full 45 comparison sensitivity family. These results describe dependence on the service objective without claiming uniform statistical evidence across all objectives. Table XVIII reports the adjusted probabilities.

Figure 2 places the primary transition comparison beside the electrical, communication, and crew execution results. The panels use the same values and sample definitions as the corresponding tables.

D. Electrical and mobility execution

Table VII summarizes the feeder and crew outcomes in the main loop. The primary run with 4,096 scenarios made 196,608 online OpenDSS calls, one for each scenario, policy, and hour. At measured load, all raw actions were already

TABLE VI

OBJECTIVE SENSITIVITY OF CENTRAL PC RELATIVE TO FORECAST MATCHED. ALL ROWS USE THE SAME 1,024 SCENARIO INPUTS. CANDIDATE ROUTES ARE SELECTED AGAIN UNDER EACH STATED OBJECTIVE. REFERENCE IS THE CORRESPONDING SUBSET OF THE PRIMARY EXPERIMENT, NOT ITS FULL 4,096 PAIRS.

Weights	Reduction (%)	Difference (95% CI)
Reference	0.0995	-1.1299 [-1.9426, -0.3834]
2× mobility	0.1028	-1.1782 [-2.0284, -0.3950]
2× unserved energy	0.0597	-1.1458 [-2.0653, -0.3006]
2× communication	0.1029	-1.3283 [-2.3910, -0.3448]
2× power service	0.1264	-1.4710 [-2.5631, -0.4754]
0.5× cascade	0.0810	-0.8553 [-1.5909, -0.1857]

TABLE VII

ELECTRICAL PROJECTION AND REPAIR COMPLETION IN THE INTEGRATED EXPERIMENTS. PRIMARY INCLUDES EIGHT POLICIES AND STRESS INCLUDES SIX. ACTION RETENTION IS THE MEAN HOURLY FRACTION. JOB COMPLETION IS THE MEAN SCENARIO FRACTION OF REQUESTED JOBS COMPLETED WITHIN THE HORIZON.

Condition	Pairs	AC hours	Raw infeas.	After proj.
Measured load	4,096	196,608	0	0
20× EV load	1,024	36,864	348	0

Condition	Action retained (%)	Jobs completed (%)
Measured load	100.00	81.05
20× EV load	99.80	81.12

feasible, so projection retained 100% of requested charging. This feasibility result does not by itself demonstrate feedback. The companion 20× stress run therefore scaled both new energy demand and the charging budget, using 1,024 paired scenarios and six policies in the same loop. It produced 348 infeasible policy hours. Projection reduced this count to zero, retained 99.80% of requested service on average, and placed 16,660.6 kWh of curtailed energy into unserved demand/backlog rather than crediting it as served. Thus the online adapter changes downstream state whenever the AC

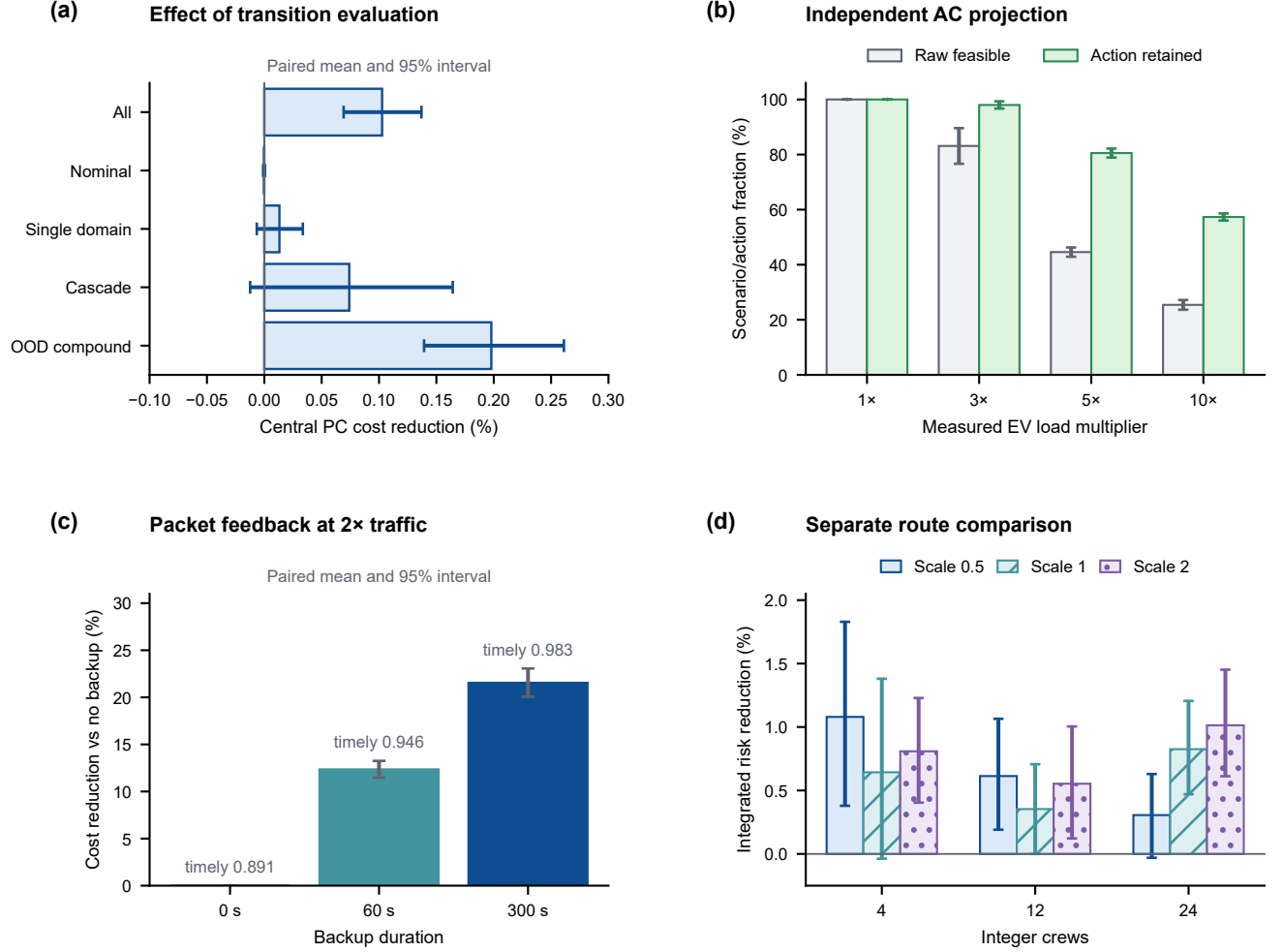


Fig. 2. Decision and execution evidence. Panel (a) reports the central PC cost reduction relative to the forecast matched baseline. Filled intervals show paired means and bootstrap confidence limits for each threat class and for all scenarios. Panel (b) reports raw distribution system feasibility and the fraction of requested EV action retained after projection, with normal approximation intervals across 20 mappings of 24 hours each. Panel (c) reports the cost reduction from backup power under increased packet traffic, with the mean fraction of actions delivered on time printed above each bar. Panel (d) reports the separate integrated risk comparison when executable routes are evaluated before selection. Panels (a), (c), and (d) show paired bootstrap 95% intervals, using 4,096 pairs overall and 1,024 per class in (a), 1,024 in (c), and 128 per crew and service condition in (d). Paired t and Wilcoxon tests with their stated Holm families are reported in the tables. Percentage intervals divide the cost or risk difference interval by the observed reference mean.

TABLE VIII
SMART-DS AC ACTION PROJECTION (20 MAPPINGS, 24 HOURS EACH)

EV load	Raw feasible (%)	Action retained (%)	Raw $V_{\min}$	Raw line p.u.
1x	100.0	100.0	0.986	0.929
3x	83.1	98.0	0.977	0.968
5x	44.6	80.5	0.966	1.044
10x	25.4	57.3	0.938	1.361

TABLE IX
STATION CHOICE UNDER MEASURED DEMAND FOR 24 PEAK HOURS

Closure (%)	No choice	Nearest	Logit	Coordinated	Extra km
10	67.4	100.0	100.0	100.0	0.32
20	55.8	100.0	100.0	100.0	0.50
30	43.7	100.0	100.0	100.0	0.97

limit binds.

Table VIII extends the electrical comparison across load multipliers. The SMART-DS experiment comprised 20 randomized station to load mappings, 24 test hours, and four EV load multipliers (1,920 cases). Raw actions were all feasible at measured load. At 3x, 5x, and 10x load, raw feasibility fell to 83.1%, 44.6%, and 25.4%, whereas the action projection retained 98.0%, 80.5%, and 57.3% of requested EV power.

Every projected case converged and met the voltage and line limits. A separate test for all of 2023 used all 7,451 nonzero hours at 1x and 3x load (14,902 cases). At 3x, 99.17% of raw actions were feasible and projection retained 99.94% on average. All projected cases remained feasible. The reported curtailment is therefore produced inside the action path and not inferred from a later voltage label.

Table IX reports the station reassignment results. Station

TABLE X

BACKUP DURATION FOR CENTRAL PC AT $2\times$ PACKET TRAFFIC ON 1,024 PAIRED SCENARIOS. COMPLETION IS THE FRACTION OF REQUESTED JOBS COMPLETED. COST DIFFERENCES COMPARE THE SAME POLICY WITH NO BACKUP.

Backup	Timely action	Completion	Reduction (%)	Difference	95% CI
0 s	0.891	0.754	0.00		Reference
60 s	0.946	0.790	12.35	[-186.05, -161.06]	
300 s	0.983	0.819	21.55	[-323.85, -281.66]	

TABLE XI

ROUTE EVALUATED PC COMPARED WITH MATCHED ROUTING

Crews	Service scale	Risk reduction (%)	Difference	95% CI	Holm p
4	0.5	1.08	[-2.25, -0.47]	0.0237	
4	1.0	0.64	[-2.33, 0.06]	0.177	
4	2.0	0.81	[-2.85, -0.94]	0.00175	
12	0.5	0.61	[-0.65, -0.12]	0.0492	
12	1.0	0.35	[-0.68, 0.00]	0.129	
12	2.0	0.55	[-1.63, -0.20]	0.00652	
24	0.5	0.31	[-0.30, 0.01]	0.0293	
24	1.0	0.83	[-0.94, -0.37]	6.49e-05	
24	2.0	1.01	[-1.95, -0.82]	6.91e-06	

closure does not automatically imply abandoned charging. A separate capacity constrained flow model distinguished local service, rerouting, and unserved demand across 24 observed peak hours. Under measured demand and 10, 20, and 30% closures, preserving the original station served only 67.4, 55.8, and 43.7% of energy. Nearest, logit, and coordinated reassignment all served 100% in these cases. The coordinated solution required only 0.32, 0.50, and 0.97 km energy weighted extra distance. Under $4\times$ demand, total constructed capacity, rather than the choice model, became the limiting factor. This stress result prevents the rerouting mechanism from creating fictitious capacity.

E. Communication and crew execution

The packet emulator is a finite buffer queue with first come, first served discipline, discrete events, erasures, two retransmissions, backoff, a 500 ms control deadline, and explicit backup duration. An M/M/1 unit test at 10 packet/s arrival and 20 packet/s service reproduced the theoretical 100 ms mean latency with 0.269% relative error. Across 12,000 station stress cases, the original algebraic support proxy had Spearman correlation 0.860 but mean absolute error 0.243 against the simulated fraction of actions delivered on time, which confirms that the proxy cannot replace packet dynamics.

Table X reports the effect of backup duration. Equation (8) closes this evidence loop. Under prespecified $2\times$ traffic and 1,024 paired uncertain transition scenarios, 60 s backup raised the mean timely action fraction to 0.946 and reduced central PC cost by 12.35% relative to no backup (difference CI [-186.0, -161.1]). A 300 s backup raised the fraction to 0.983 and reduced cost by 21.55% (difference CI [-323.8, -281.7]). Both paired mean tests remained significant after Holm correction across the two backup comparisons. Before backup exhaustion, power threat caused no direct service rate derating. The effect arose only after the coded depletion event.

In the integrated primary run, the route generated for four crews was not a diagnostic added after scoring. The same work order set was dispatched through packet delivery, and threat was updated only at its recorded completion hours. Central PC completed a mean 81.20% of requested jobs within the horizon versus 81.10% for forecast matched. The fraction is calculated separately in each scenario before averaging, and its denominator includes requests not dispatched. The paired difference interval was [0.00058, 0.00147]. Mean scheduled completion time among dispatched jobs fell from 3.119 to 3.097 h, with difference interval [-0.02643, -0.01802] h. This time includes jobs whose scheduled completion extends beyond the service horizon. Total scheduled travel was 2.621 versus 2.619 h per scenario, with difference interval [-0.00358, 0.00765] h. The completion fraction and time comparisons remained significant after correction across nine secondary service and physical endpoints, whereas travel did not improve detectably. Table XVI reports all nine outcomes. Unserved energy decreased by 0.0369 kWh per scenario, while electrical loss increased by 1.53×10^{-6} MWh, with interval $[0.60, 2.46] \times 10^{-6}$ MWh. Both electrical changes were small. The voltage and timely action mean intervals included zero, and neither action retention nor curtailment differed at measured load. Loss and voltage measurements used separate OpenDSS contexts at convergence tolerance 10^{-10} , compared with 10^{-8} , and at most 100 iterations. The contexts received identical accepted loads for all 4,096 central and matched pairs without changing scores, routes, executed service, or the original 10^{-4} execution tolerance. The largest scenario differences between measurement tolerances were 9.94×10^{-8} MWh and 6.35×10^{-10} p.u., respectively.

Table XI reports the route comparison for every crew and service condition. The route experiment used 36 jobs, integer fleets of 4, 12, and 24 crews, geocoded travel at 30 km/h with 1.25 circuitry, and service time multipliers 0.5, 1, and 2. Each repair completion causally removed that job's risk from the accumulated state. The original PC priority lost its advantage in some integer routed cases. Route evaluated PC corrected this result by selecting the lowest predicted risk executable candidate rather than projecting a continuous priority after the decision. In 128 independent scenarios per cell, route evaluated PC reduced mean integrated risk in all nine crew and service conditions by 0.31% through 1.08%. Six bootstrap intervals excluded zero and seven Holm adjusted Wilcoxon tests were below 0.05. The intervals for four crews at scale 1.0, twelve crews at scale 1.0, and twenty four crews at scale 0.5 included zero and remain visible in Table XI. The observed gain is not declared uniformly significant. The Holm adjusted Wilcoxon value for twenty four crews at scale 0.5 is below 0.05 while its bootstrap interval crosses zero because the two procedures test different statistics. The interval targets the paired mean difference, whereas Wilcoxon ranks the nonzero paired differences and is affected differently by ties and skewness. We therefore require the mean interval, not either test in isolation, for a mean improvement claim.

The integrated sensitivity study holds the realized coefficient distribution fixed while perturbing the matrix available to the controller. Each condition uses the same 1,024

scenario inputs. With $0.75M$, central PC exceeded matched cost by 3.7779, with marginal interval $[2.4720, 5.0919]$, whereas robust PC was 3.6909 below central PC with interval $[-4.7773, -2.6813]$. With $1.25M$, central PC reduced matched cost by 1.2737, with interval $[-2.0959, -0.5635]$. Under coefficient noise of 0.15, the difference was -1.8341 with interval $[-2.6349, -1.1321]$. Robust PC exceeded central PC by 3.4569 and 1.2651 in these latter conditions. Thus worst set scoring offset the underscaled transition in one condition but did not provide uniform protection. The calibrated transition's primary advantage also did not persist under every matrix error. Table XVIII gives these comparisons with the full sensitivity family and its multiplicity correction.

To distinguish uncertainty in the controller from uncertainty in the realized environment, additional runs changed only the realized coefficients. The communication persistence and four couplings without joint field estimates were set to zero individually and together, or doubled together, while the controller retained its calibrated matrix and declared uncertainty set. Setting all five quantities to zero gave a central minus matched difference of -1.1745 , with marginal interval $[-1.7176, -0.6815]$, a 0.1156% reduction. The individual zero settings also had negative mean differences, although this alone does not establish significance across the full comparison family. Doubling the five quantities gave 0.2837 with interval $[-0.7188, 1.2761]$, so a mean benefit was not established in that condition. Robust PC was then 2.0608 below central PC, with interval $[-2.9543, -1.2204]$.

A separate test changed the form of the realized transition rather than its linear coefficients alone. With $A_{sp} = \text{diag}(A, A, A)$, the propagated state before innovations and completion was

$$T(\zeta) = 0.98 \left[1 - \exp \left(- \frac{[M_{\text{real}}\zeta]_+ + 0.15(A_{sp}\zeta)^{\odot 2}}{0.98} \right) \right], \quad (81)$$

where the exponential and square act componentwise. Execution then used $D(\bar{r}_h)\Pi_{\mathcal{Z}}(T(\zeta_h) + \xi_h)$. The quadratic spatial term and saturation in (81) are absent from the controller's linear matrix family. On 1,024 paired scenarios, central PC reduced matched cost by 0.1532%, with difference -1.5148 and interval $[-1.9601, -1.1040]$. Robust PC exceeded central PC by 0.7813, with interval $[0.4315, 1.1492]$. These results test a distinct transition form without changing the scoring rule or granting future information to the policy. Table XVIII retains every zero coupling, expanded coupling, and nonlinear comparison alongside the matrix error tests.

F. Numerical and mathematical checks

The continuous service cost and the executor were compared in 24 cases with packet and electrical factors fixed to one, identical forecasts, identical transitions, and the same completion schedule. The largest absolute cost difference was 2.44×10^{-4} from floating point arithmetic. Changing future observed demand did not alter the candidate score, and a missing forecast used only the preceding observation. Target windows respected the training and validation boundaries. Additional Jacobian checks examined the LayerNorm bound at

constant, nearly constant, and dispersed inputs and the GELU derivative bound. These finite checks support correspondence with the formulas, while the all input statements on the declared domain follow from the proofs.

Route error was examined by evaluating every ordered assignment of at most four dispatched jobs to two interchangeable crews in 64 small scenarios, and all six primary candidates in another 128 scenarios. The two tests required 3,678 candidate evaluations. Mean realized regret of service scoring was 0.0028 in the small exhaustive test and 2.2108 in the primary portfolio test, compared with 0.0080 and 17.0115 for exposure scoring on the same candidates. The service selector reached the realized candidate optimum in 92.19% and 62.50% of cases, respectively. Every instance satisfied both error bounds in Theorem 2. Absolute score errors can include a large common demand prediction offset that does not change route ordering, which explains the greater relevance of the centered error range. These are observed errors on finite candidate sets, not an estimated uniform error guarantee for new scenarios.

Electrical replay checks covered six load scales, including three infeasible requests. The accepted fraction was reapplied and solved again before its service was credited. Each returned action matched the solved load, and a constructed collapse of a base active node remained visible to the voltage test. All 14,902 full year checks, 196,608 primary policy hours, and 36,864 stress policy hours were feasible after projection. These execution checks test the numerical requirements of Proposition 1 without extending the continuous Lipschitz conclusion to discrete electrical or repair events. Independent electrical measurements covered another 49,152 primary policy hours at each of two solver tolerances. All actions remained feasible and every cost, service, route, and completion outcome was unchanged. Repeating 32 paired scenarios with reversed policy order changed the 10^{-10} loss measurements by at most 3.10×10^{-8} MWh per scenario. This precision check supports reporting the small loss increase without attributing it to the original solver tolerance.

G. Interpretation and reproducibility

The matched comparisons separate two contributions to route selection. Service costs and backlog account for the larger gain over exposure scoring. Propagation adds a smaller average benefit when the service objective and available information are held fixed. The overall and compound stress mean intervals support this increment, but the remaining classwise intervals include zero.

The robust extension provides a separate comparison with central PC. Its primary mean and conditional tail costs are higher, and its maximum cost is unchanged. Its protection under an underscaled transition does not extend to all tested errors in the controller's matrix. The scoring choices in Table IV therefore distinguish service cost reduction, propagation benefit, and conditional protection against model error.

Execution results explain the route from a requested decision to a realized state. Packet delivery, feeder acceptance, and crew completion determine whether a request produces service

or repair. The feeder constraint is inactive at measured load but changes service and backlog under electrical stress. This independent feasibility calculation checks execution, whereas relative cost benefits remain estimates under the modeled service objective and transition scenarios.

The three public data sources enter through distinct variables. Boulder supplies charging demand, EAGLE-I supplies outage timing, and SMART-DS supplies the electrical network, as specified in Table I. Their combination defines a service restoration experiment, not observations of a single coincident infrastructure. Constructed communication conditions and repair durations connect these inputs through explicit execution rules.

The materials supporting these results are available in a public repository. It contains code, processed inputs, trained model parameters, the distribution feeder, and scenario records for regenerating the reported statistics. Data documentation specifies public sources, licenses, variable definitions, time periods, preprocessing, and energy conservation checks. The software environment, numerical settings, and execution records are also provided at <https://github.com/ahshicr/TSG-01697-2026-reproducibility>.

V. CONCLUSION

This paper develops restoration decisions that evaluate repair routes through predicted service costs, backlog, and calibrated threat propagation.¹ Packet delivery, electrical feasibility, and integer repair completion connect these decisions to realized service. The analysis establishes continuous sensitivity bounds, conditional finite route regret, and execution constraints. Matched comparisons distinguish the benefit of service cost scoring from the smaller average contribution of propagation, most clearly observed under compound stress. Worst set scoring provides protection in selected error conditions but does not improve primary mean or tail costs. These results support a service restoration model with explicit execution rules and comparisons that separate the effects of the objective and transition model.

APPENDIX

PARAMETERS AND ADDITIONAL COMPARISONS

Tables XII, XIII, and XIV state the numerical settings, their roles, sources, selection periods, and tested ranges.¹ Grouped values follow the listed parameter order. Unless a data source or sensitivity range is stated, values are declared benchmark settings held fixed across policies. They were not retuned on the revised test. Table III separately gives transition estimates and intervals. Long estimates are rounded for display.

Table XV retains the complete primary comparison family. Table XVIII gives every integrated sensitivity comparison, Table XVI gives all secondary service and physical outcomes, and Table XVII gives the empirical cost tails. The numerical record retains all paired mean and rank tests, including unfavorable comparisons.

TABLE XII
SCENARIO CONSTRUCTION AND STATISTICAL SETTINGS.

Parameter	Value	Unit, role, source, selection and sensitivity
Affected zones, severity, innovation	(2,6,8,12), (0.08, 0.35,0.45,0.62), (0.003,0.008, 0.012,0.022)	Values follow nominal, single domain, cascade and compound order within each sequence. Nominal affects road only, single domain draws one of the three domains uniformly, and the remaining classes affect all three.
Initial and innovation multipliers	0.75 to 1.25, 0.6 to 1.4	Independent uniform draws over the displayed intervals. Nominal innovations affect only the first sampled location. Other classes perturb every sampled location in each affected domain.
Location sampling offset	0.05	Sampling probability is proportional to the training mean arrival count plus this fraction of its spatial mean. Locations are sampled without replacement.
Initial and propagated state caps	0.95, 0.98	Normalized upper limits. Both initial and propagated states are clipped below by zero. The second value is also the nonlinear saturation scale.
Primary pairs, resamples, primary tests, physical tests	4096, 20000, 15, 9	Four equal threat groups, percentile 95 percent intervals and separate Holm adjustments for paired mean and rank tests. Integrated sensitivity has 45 comparisons.
Initial index, scenario stride, noise index	2026, 31991, 7919	A scenario uses the initial index plus its identifier times the stride. The last index fixes controller coefficient noise. The paired policies share the resulting inputs. No index was retuned.

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TABLE XIII
PREDICTION, OBJECTIVE, SERVICE AND ALLOCATION SETTINGS.

Parameter	Value	Unit, role, source, selection and sensitivity
History L , horizon H	12, 6	Observed and predicted hours.
Hidden units, epochs, batch	32, 40, 256	Hidden width, maximum epochs and batch samples. Checkpoint selected by validation error.
Learning rate, decay, dropout	0.002, 0.0001, 0.1	AdamW rate, weight decay and dropout fraction, shared by all chronological fits.
ϵ_{LN} , Huber threshold, gradient clip	10^{-5} , 1.0, 2.0	Variance stabilizer, normalized loss threshold and Euclidean gradient bound.
Training repetitions, neighbours	3, 4	Independent fits and geographical neighbours. Training uses 2018 to 2021, selection uses 2022. Graph uses station coordinates.
Exposure weights, bounds, mean floor	(0.25,0.25), 1 to 6, 10^{-6}	Arrival and energy weights, clipping interval and positive training mean floor in the corresponding arrival or energy units.
Discount γ	1.0	Undiscounted six hour score.
$\lambda_m, \lambda_e, \lambda_c, \lambda_p, \lambda_z$	1.8, 2.6, 1.7, 1.2, 12.0	Reciprocal service units defined in the objective. Sensitivities use 3.6, 5.2, 3.4, 2.4 and 6, one weight at a time.
$\beta_m, \beta_e, \beta_r, \beta_p$	0.13, 0.19, 6.0, 3.5	Mobility, energy, road threat and power threat conversions to communication load.
$\alpha_c, \beta_{mc}, \rho_q$	0.40, 0.45, 0.22	Communication derating, communication shortfall to mobility loss and backlog carry-over. Carryover is below one.
Communication shortfall cap	0.90	Maximum fractional reduction of charging support.
$\kappa_q, \kappa_{ch}, \kappa_E, \kappa_{com}$	1.0, 1.5, 0.7, 1.5	Backlog and threat weights in charging, then energy and threat weights in communication.
$\omega_p, \omega_c, \omega_r, \omega_\Gamma$	1.6, 1.3, 1.1, 0.8	Power, communication, road and predicted marginal benefit weights in repair priority.
Service reserve, repair reserve, ϵ	0.04, 0.02, 10^{-9}	Uniform reserve fractions and positive score stabilizer.
Charging budget B^{ch}	18.48321	kWh per hour. Training station mean energy sum multiplied by 1.22. Electrical stress uses factor 24.4 with twenty times demand.
Communication budget B^{com}	73.144081	Service units. Training capacity proxy sum multiplied by 0.72. The proxy uses training arrivals and connected session hours.
Repair priority budget B^{res}	10.0	Priority units. Maximum of 10 and 0.018 times summed training mean arrivals. All three budgets use 2018 to 2021 only.
Road, power, communication repair effect	0.08, 0.12, 0.10	Threat reduction per quantum in continuous priority prediction, not realized repair.
Predictive repair quantum cap	3.0	Priority divided by the per zone reference, clipped from zero to this cap.

TABLE XIV
CREW, COMMUNICATION, ELECTRICAL AND STATION SETTINGS.

Parameter	Value	Unit, role, source, selection and sensitivity
Crews, maximum jobs	4, 36	Crews and work orders. Sensitivity uses 12 and 24 crews. Job count follows initial threat.
Travel speed, circuitry	30, 1.25	km per hour and road to great circle distance ratio at Boulder coordinates.
Repair duration	empirical	Hours. EAGLE-I 50 customer half recovery prior, 255 observations among 1,467 events ending before 2022, clipped to 0.25 to 8 hours.
Spatial job weights, inclusion threshold	(0.60,0.30), 10^{-10}	One and two neighbour contributions to initial threat, then the inclusion threshold. Jobs are fixed before policy selection.
Scaled service duration interval	0.25 to 16	Hours after scaling the base prior by 0.5, 1 or 2, then clipping.
Route construction	deterministic list	Priority per incremental completion time, fixed ties and no time cutoff. Common candidates for all selectors.
Queue window, arrival rate, service rate	120, 8+0.8 activity, 30	Seconds and packets per second. Activity is connected sessions plus arrivals from twenty training hours. Traffic factors are 0.5, 1, 2 and 4.
Buffer, control share, deadline	100, 0.05, 500	Packets, fraction and milliseconds.
Retries, mean backoff	2, 0.05	Maximum retransmissions and exponential retry wait in seconds.
Erasure probability, backup	$\min(0.25, 0.002 + 0.08z^c)$, 60	Threat dependent erasure and backup in seconds. Sensitivity uses 0 and 300 seconds.
Queue derating, rate floor, context ratio	0.70, 0.75, 0.5, 0.5	Base rate factor $1 - 0.70z^c$, multiplied by $1 - 0.75z^p$ after backup. Rate floor in packets per second. Constructed $z^c/z^p = 0.5$.
Power and communication lookup scales	0.75, 0.375	Distance scales for nearest training response selection.
Feeder, power factor	SMART-DS SFO P1U, 0.98	Complete public feeder and lagging EV power factor.
EV load model, voltage band, line limit	constant PQ model 1, 0.95 to 1.05, 1.0	Balanced three phase wye increments at 17 loads. Voltage and loading in p.u. Base loads unchanged.
Bisection steps, execution tolerance	14, 10^{-4}	Fraction resolution about 0.000061. Original OpenDSS convergence criterion unchanged.
Measurement tolerance, iteration limit	10^{-10} , 100	Independent measurements for all central and matched pairs. Second tolerance is 0.00000001. Accepted actions unchanged.
Mappings, initialization index	20, 20260825	Station to load assignments. Scenario index modulo twenty selects the shared assignment.
Station capacity	$\max(1.2q_{0.995}, 7.2)$	kWh per hour. Training energy quantile at probability 0.995. Stress proxy, not nameplate power.
Closure, demand, distance, loyalty	(0.1,0.2,0.3), (1,2,4), (0.5,1,2), 2.0	Fraction, multiplier, reciprocal km and cost units. Full factorial at 24 selected 2023 hours.

TABLE XV

CENTRAL PC AGAINST BOTH SERVICE AND EXPOSURE SCORE CONTROLS. HOLM CORRECTION INCLUDES ALL FIFTEEN COMPARISONS SEPARATELY FOR EACH TEST. BOOTSTRAP INTERVALS ARE MARGINAL INTERVALS RATHER THAN SIMULTANEOUS INTERVALS.

Comparator	Group	Pairs	Δ	95% CI	Holm t	Holm Wilcoxon
Forecast matched	All	4096	-1.1436	[-1.5257, -0.7704]	1.38×10^{-8}	9.79×10^{-23}
Forecast matched	Nominal	1024	0.0013	[-0.0042, 0.0075]	1.0000	3.77×10^{-17}
Forecast matched	Single domain	1024	-0.1021	[-0.2602, 0.0510]	0.5860	0.0020
Forecast matched	Cascade	1024	-0.7946	[-1.7624, 0.1327]	0.3944	0.0983
Forecast matched	OOD compound	1024	-3.6789	[-4.8554, -2.5890]	1.94×10^{-9}	1.15×10^{-11}
Exposure matched	All	4096	-7.6977	[-8.3843, -7.0077]	2.83×10^{-97}	5.2×10^{-147}
Exposure matched	Nominal	1024	0.0019	[-0.0037, 0.0081]	1.0000	7.68×10^{-16}
Exposure matched	Single domain	1024	-0.2304	[-0.4169, -0.0533]	0.0626	3.11×10^{-6}
Exposure matched	Cascade	1024	-8.3878	[-9.6902, -7.1245]	2.38×10^{-34}	1.39×10^{-44}
Exposure matched	OOD compound	1024	-22.1746	[-24.3945, -20.0079]	6.17×10^{-72}	1.96×10^{-76}
Exposure central	All	4096	-12.0931	[-13.1616, -11.0704]	1.98×10^{-104}	1.32×10^{-139}
Exposure central	Nominal	1024	0.0103	[0.0047, 0.0165]	0.0036	0.4691
Exposure central	Single domain	1024	-0.3792	[-0.5915, -0.1773]	0.0031	0.1441
Exposure central	Cascade	1024	-8.5773	[-9.9791, -7.2295]	4.84×10^{-32}	2.28×10^{-40}
Exposure central	OOD compound	1024	-39.4260	[-42.9641, -36.0140]	5.55×10^{-89}	1.29×10^{-95}

TABLE XVI

PHYSICAL OUTCOMES OF CENTRAL PC (C) AND FORECAST MATCHED (F) ON 4,096 PAIRED SCENARIOS. INTERVALS CONCERN C MINUS F. HOLM CORRECTION COVERS ALL NINE ENDPOINTS SEPARATELY FOR EACH TEST. COMPLETION FRACTION USES REQUESTED JOBS, INCLUDING UNDELIVERED REQUESTS. SCHEDULED COMPLETION AND TRAVEL INCLUDE DISPATCHED JOBS FINISHING BEYOND THE HORIZON. DECISION COMPUTATION TIME IS EXCLUDED BECAUSE CONCURRENT EXECUTION IS NOT A CONTROLLED TIMING COMPARISON. ELECTRICAL LOSS AND MINIMUM VOLTAGE USE INDEPENDENT OPENDSS MEASUREMENTS OF IDENTICAL ACCEPTED LOADS AT CONVERGENCE TOLERANCE 10^{-10} , WITH 10^{-8} COMPARISON SOLVES AND AT MOST 100 ITERATIONS. THESE MEASUREMENTS DO NOT ALTER REQUESTS, ACCEPTED ACTIONS, ROUTES, OR COSTS.

Endpoint	C mean	F mean	Difference	95% CI	Holm t	Holm Wilcoxon
Unserved energy (kWh)	294.358909	294.395760		[-0.057777, -0.015918]	0.0037	1.26×10^{-12}
Electrical loss (MWh)	0.841670840	0.841669314		$[6.013 \times 10^{-7}, 2.457 \times 10^{-6}]$	0.0071	2×10^{-14}
Minimum voltage (p.u.)	0.989098473	0.989098647		$[-4.103 \times 10^{-7}, 5.085 \times 10^{-8}]$	0.7135	0.0117
Timely action fraction	0.976412	0.976401		$[-2.077 \times 10^{-5}, 4.310 \times 10^{-5}]$	1.0000	1.0000
Retained action fraction	1.000000	1.000000		[0, 0]	1.0000	1.0000
Curtailment (kWh)	0	0		[0, 0]	1.0000	1.0000
Completed/requested jobs	0.811998	0.810978		$[5.779 \times 10^{-4}, 0.001470]$	6.51×10^{-5}	0.0013
Scheduled completion (h)	3.096674	3.118857		[-0.026427, -0.018020]	2.11×10^{-23}	8.59×10^{-18}
Scheduled travel (h)	2.621301	2.619267		[-0.003585, 0.007646]	1.0000	1.0000

TABLE XVII

PRIMARY EMPIRICAL COST DISTRIBUTION. QUANTILES AND UPPER TAIL MEANS DESCRIBE THE SAME 4,096 SCENARIOS PER POLICY. TAIL MEANS INCLUDE A FRACTIONAL OBSERVATION AT THEIR PROBABILITY BOUNDARY. THESE DESCRIPTIVE VALUES ARE NOT INFERENTIAL EVIDENCE OF TAIL SUPERIORITY.

Policy	Mean	95% quantile	99% quantile	Upper 5% mean	Upper 1% mean	Maximum
Forecast matched	1113.68	2599.56	3334.41	3062.58	3694.88	4494.26
Central PC	1112.54	2597.87	3331.67	3055.50	3687.60	4494.26
Robust PC	1112.90	2603.73	3331.44	3058.35	3692.33	4494.26
Exposure matched	1120.23	2618.63	3357.45	3082.80	3729.44	4531.67
Exposure central	1124.63	2632.39	3369.16	3105.46	3746.92	4507.25
Static	1202.74	2767.64	3507.76	3224.38	3895.21	4799.52
Greedy	1122.54	2632.59	3372.35	3093.79	3744.89	4494.26
Oracle	1053.67	2582.47	3301.28	3019.15	3647.82	4474.21

TABLE XVIII

ALL INTEGRATED SENSITIVITY COMPARISONS. C DENOTES CENTRAL PC, F FORECAST MATCHED, AND R ROBUST PC. THE MEAN DIFFERENCE IS FIRST POLICY MINUS SECOND POLICY. HOLM CORRECTION INCLUDES THE FULL SENSITIVITY FAMILY. INTERVALS ARE MARGINAL.

Condition	Comparison	Pairs	Δ	95% CI	Holm t	Holm Wilcoxon
Primary	R minus C	4096	0.3670	[0.0788, 0.6627]	0.2589	0.1847
20× EV load	C minus F	1024	-3.7920	[-9.9833, 2.2046]	1.0000	1.0000
20× EV load	R minus C	1024	-0.6276	[-6.5897, 5.3242]	1.0000	1.0000
No backup	C minus F	1024	-1.7641	[-2.1646, -1.3886]	2.04×10^{-16}	3.66×10^{-35}
No backup	R minus C	1024	0.1626	[-0.0448, 0.4040]	1.0000	1.0000
60 s backup	C minus F	1024	-1.0259	[-1.6009, -0.4900]	0.0086	7.19×10^{-12}
60 s backup	R minus C	1024	0.2204	[-0.1668, 0.6233]	1.0000	1.0000
300 s backup	C minus F	1024	-1.9467	[-2.8292, -1.0894]	0.0005	1.36×10^{-10}
300 s backup	R minus C	1024	0.8154	[0.0944, 1.5804]	0.5204	0.1050
0.75M	C minus F	1024	3.7779	[2.4720, 5.0919]	8.54×10^{-7}	0.0002
0.75M	R minus C	1024	-3.6909	[-4.7773, -2.6813]	4.01×10^{-10}	9.13×10^{-14}
1.25M	C minus F	1024	-1.2737	[-2.0959, -0.5635]	0.0324	7.69×10^{-7}
1.25M	R minus C	1024	3.4569	[2.4826, 4.5032]	1.02×10^{-9}	4.68×10^{-11}
Coefficient noise 0.15	C minus F	1024	-1.8341	[-2.6349, -1.1321]	6.76×10^{-5}	2.13×10^{-14}
Coefficient noise 0.15	R minus C	1024	1.2651	[0.6919, 1.8547]	0.0006	1.52×10^{-5}
12 crews	C minus F	1024	-0.3849	[-0.4717, -0.3068]	8.38×10^{-18}	1.02×10^{-52}
12 crews	R minus C	1024	0.0279	[-0.0273, 0.0874]	1.0000	1.0000
24 crews	C minus F	1024	-0.6119	[-0.7337, -0.5022]	7.53×10^{-23}	1.49×10^{-74}
24 crews	R minus C	1024	-0.0236	[-0.0538, -0.0003]	1.0000	0.0934
$\rho_c = 0$	C minus F	1024	-1.4042	[-2.0789, -0.7983]	0.0006	8.37×10^{-10}
$\rho_c = 0$	R minus C	1024	0.7161	[0.2249, 1.2241]	0.1195	0.0311
$\tau_{pc} = 0$	C minus F	1024	-1.5121	[-2.1857, -0.9005]	0.0002	3.08×10^{-11}
$\tau_{pc} = 0$	R minus C	1024	0.7820	[0.2952, 1.2915]	0.0562	0.0063
$\tau_{rc} = 0$	C minus F	1024	-1.1054	[-1.9120, -0.3671]	0.1144	0.0007
$\tau_{rc} = 0$	R minus C	1024	0.2731	[-0.3154, 0.8676]	1.0000	1.0000
$\tau_{cr} = 0$	C minus F	1024	-1.4150	[-2.1974, -0.7115]	0.0052	1.18×10^{-7}
$\tau_{cr} = 0$	R minus C	1024	0.5521	[-0.0121, 1.1336]	0.9031	0.4140
$\tau_{cp} = 0$	C minus F	1024	-0.9430	[-1.6911, -0.2523]	0.2088	0.0005
$\tau_{cp} = 0$	R minus C	1024	0.3110	[-0.2648, 0.8932]	1.0000	1.0000
All constructed couplings zero	C minus F	1024	-1.1745	[-1.7176, -0.6815]	0.0003	1.02×10^{-9}
All constructed couplings zero	R minus C	1024	0.7285	[0.2995, 1.1792]	0.0326	0.0009
Constructed couplings doubled	C minus F	1024	0.2837	[-0.7188, 1.2761]	1.0000	1.0000
Constructed couplings doubled	R minus C	1024	-2.0608	[-2.9543, -1.2204]	0.0001	2.23×10^{-6}
Nonlinear realized transition	C minus F	1024	-1.5148	[-1.9601, -1.1040]	3.75×10^{-10}	4.53×10^{-19}
Nonlinear realized transition	R minus C	1024	0.7813	[0.4315, 1.1492]	0.0006	4.6×10^{-5}
2× mobility	C minus F	1024	-1.1782	[-2.0284, -0.3950]	0.1092	0.0006
2× mobility	R minus C	1024	0.3137	[-0.3283, 0.9612]	1.0000	1.0000
2× unserved energy	C minus F	1024	-1.1458	[-2.0653, -0.3006]	0.2088	0.0006
2× unserved energy	R minus C	1024	0.1823	[-0.4827, 0.8508]	1.0000	1.0000
2× communication	C minus F	1024	-1.3283	[-2.3910, -0.3448]	0.2088	0.0003
2× communication	R minus C	1024	0.2559	[-0.5376, 1.0534]	1.0000	1.0000
2× power service	C minus F	1024	-1.4710	[-2.5631, -0.4754]	0.1198	0.0002
2× power service	R minus C	1024	0.3342	[-0.4771, 1.1526]	1.0000	1.0000
0.5× cascade	C minus F	1024	-0.8553	[-1.5909, -0.1857]	0.2760	0.0011
0.5× cascade	R minus C	1024	0.1604	[-0.3922, 0.7155]	1.0000	1.0000

REFERENCES

- [1] F. He, Y. Yin, and J. Zhou, "Deploying public charging stations for electric vehicles on urban road networks," *Transportation Research Part C: Emerging Technologies*, vol. 60, pp. 227–240, 2015.
- [2] W. Wei, S. Mei, L. Wu, M. Shahidehpour, and Y. Fang, "Optimal traffic-power flow in urban electrified transportation networks," *IEEE Transactions on Smart Grid*, vol. 8, no. 1, pp. 84–95, 2017.
- [3] C. Shao, T. Qian, Y. Wang, and X. Wang, "Coordinated planning of extreme fast charging stations and power distribution networks considering on-site storage," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 1, pp. 493–504, 2021.
- [4] Y. Cui, Z. Hu, and X. Duan, "Optimal pricing of public electric vehicle charging stations considering operations of coupled transportation and power systems," *IEEE Transactions on Smart Grid*, vol. 12, no. 4, pp. 3278–3288, 2021.
- [5] K. Li, C. Shao, X. Li, Z. Hu, M. Shahidehpour, and X. Wang, "An inverse optimization method for electric vehicle charging and road toll pricing in coupled urban transportation and power distribution systems," *IEEE Transactions on Smart Grid*, vol. 15, no. 2, pp. 1849–1860, 2024.
- [6] L. Ran, J. Qin, Y. Wan, W. Fu, W. Yu, and F. Xiao, "Fast charging navigation strategy of evs in power-transportation networks: A coupled network weighted pricing perspective," *IEEE Transactions on Smart Grid*, vol. 15, no. 4, pp. 3864–3875, 2024.
- [7] S. M. Rinaldi, J. P. Peerenboom, and T. K. Kelly, "Identifying, understanding, and analyzing critical infrastructure interdependencies," *IEEE Control Systems Magazine*, vol. 21, no. 6, pp. 11–25, 2001.
- [8] M. Ouyang, "Review on modeling and simulation of interdependent critical infrastructure systems," *Reliability Engineering and System Safety*, vol. 121, pp. 43–60, 2014.
- [9] M. Panteli and P. Mancarella, "The grid: Stronger, bigger, smarter? presenting a conceptual framework of power system resilience," *IEEE Power and Energy Magazine*, vol. 13, no. 3, pp. 58–66, 2015.
- [10] B. Ti, G. Li, M. Zhou, and J. Wang, "Resilience assessment and improvement for cyber-physical power systems under typhoon disasters," *IEEE Transactions on Smart Grid*, vol. 13, no. 1, pp. 783–794, 2022.
- [11] X. Yin, G. Wu, J. Wei, Y. Shen, H. Qi, and B. Yin, "Deep learning on traffic prediction: Methods, analysis, and future directions," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 6, pp. 4927–4943, 2022.
- [12] W. Jiang and J. Luo, "Graph neural network for traffic forecasting: A survey," *Expert Systems with Applications*, vol. 207, p. 117921, 2022.
- [13] G. Jin, F. Li, J. Zhang, M. Wang, and J. Huang, "Automated dilated spatio-temporal synchronous graph modeling for traffic prediction," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 8, pp. 8820–8830, 2023.
- [14] M. Li, Y. Tang, and W. Ma, "Few-sample traffic prediction with graph networks using locale as relational inductive biases," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 2, pp. 1894–1908, 2023.
- [15] Y. Shin and Y. Yoon, "Pgcnn: Progressive graph convolutional networks for spatial-temporal traffic forecasting," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 7, pp. 7633–7644, 2024.
- [16] F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, "Digital twin in industry: State-of-the-art," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 4, pp. 2405–2415, 2019.
- [17] A. Fuller, Z. Fan, C. Day, and C. Barlow, "Digital twin: Enabling technologies, challenges and open research," *IEEE Access*, vol. 8, pp. 108 952–108 971, 2020.
- [18] R. Minerva, G. M. Lee, and N. Crespi, "Digital twin in the iot context: A survey on technical features, scenarios, and architectural models," *Proceedings of the IEEE*, vol. 108, no. 10, pp. 1785–1824, 2020.
- [19] P. Sadeghi-Barzani, A. Rajabi-Ghahnavieh, and H. Kazemi-Karegar, "Optimal fast charging station placing and sizing," *Applied Energy*, vol. 125, pp. 289–299, 2014.
- [20] M. Alizadeh, A. Scaglione, J. Davies, and K. S. Kurani, "A scalable stochastic model for the electricity demand of electric and plug-in hybrid vehicles," *IEEE Transactions on Smart Grid*, vol. 5, no. 2, pp. 848–860, 2014.
- [21] S. Deb, K. Tammi, X.-Z. Gao, K. Kalita, P. Mahanta, and S. Cross, "A robust two-stage planning model for the charging station placement problem considering road traffic uncertainty," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 7, pp. 6571–6585, 2022.
- [22] K. Li, C. Shao, Z. Hu, and M. Shahidehpour, "An milp method for optimal planning of electric vehicle charging stations in coordinated urban power and transportation networks," *IEEE Transactions on Power Systems*, vol. 38, no. 6, pp. 5406–5419, 2023.
- [23] S. Lv, Z. Wei, G. Sun, S. Chen, and H. Zang, "Optimal power and semi-dynamic traffic flow in urban electrified transportation networks," *IEEE Transactions on Smart Grid*, vol. 11, no. 3, pp. 1854–1865, 2020.
- [24] Y. Tao, J. Qiu, S. Lai, X. Sun, H. Liu, and J. Zhao, "Distributed electric vehicle assignment and charging navigation in cyber-physical systems," *IEEE Transactions on Smart Grid*, vol. 15, no. 2, pp. 1861–1875, 2024.
- [25] S. V. Buldyrev, R. Parshani, G. Paul, H. E. Stanley, and S. Havlin, "Catastrophic cascade of failures in interdependent networks," *Nature*, vol. 464, no. 7291, pp. 1025–1028, 2010.
- [26] C. Nowzari, V. M. Preciado, and G. J. Pappas, "Analysis and control of epidemics: A survey of spreading processes on complex networks," *IEEE Control Systems Magazine*, vol. 36, no. 1, pp. 26–46, 2016.
- [27] A. Arab, A. Khodaei, Z. Han, and S. K. Khator, "Proactive recovery of electric power assets for resiliency enhancement," *IEEE Access*, vol. 3, pp. 99–109, 2015.
- [28] A. Haydari and Y. Yilmaz, "Deep reinforcement learning for intelligent transportation systems: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 1, pp. 11–32, 2022.
- [29] Z. Zhang, M. Sun, R. Deng, C. Kang, and M.-Y. Chow, "Physics-constrained robustness evaluation of intelligent security assessment for power systems," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 872–884, 2023.
- [30] Z. Zhang, Z. Yang, D. K. Y. Yau, Y. Tian, and J. Ma, "Data security of machine learning applied in low-carbon smart grid: A formal model for the physics-constrained robustness," *Applied Energy*, vol. 347, p. 121405, 2023.
- [31] Z. Zhang, W. Wang, M. Wang, H. Wang, J. Bi, and G. Xu, "Physics-constrained adversarial attack generation and active defense for the hybrid deep reinforcement learning-based load frequency control," *ACM Transactions on Autonomous and Adaptive Systems*, 2026, advance online publication, March 28, 2026.
- [32] K. Ding, F. T. S. Chan, X. Zhang, G. Zhou, and F. Zhang, "Defining a digital twin-based cyber-physical production system for autonomous manufacturing in smart shop floors," *International Journal of Production Research*, vol. 57, no. 20, pp. 6315–6334, 2019.
- [33] S. Rahmani, A. Baghbani, N. Bouguila, and Z. Patterson, "Graph neural networks for intelligent transportation systems: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 8, pp. 8846–8885, 2023.
- [34] C. Guo, C.-H. Chen, F.-J. Hwang, C.-C. Chang, and C.-C. Chang, "Fast spatiotemporal learning framework for traffic flow forecasting," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 8, pp. 8606–8616, 2023.
- [35] J. Ji, F. Yu, and M. Lei, "Self-supervised spatiotemporal graph neural networks with self-distillation for traffic prediction," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 2, pp. 1580–1593, 2023.
- [36] A. Märtz, U. Langenmayr, S. Ried, K. Seddig, and P. Jochem, "Charging behavior of electric vehicles: Temporal clustering based on real-world data," *Energies*, vol. 15, no. 18, p. 6575, 2022.
- [37] C. Brelsford, S. Tennille, A. Myers, S. Chinthavali, V. Tansakul, M. Denman, M. Coletti, J. Grant, S. Lee, K. Allen, E. Johnson, J. Huihui, A. Hamaker, S. Newby, K. Medlen, D. Maguire, C. Dunivan Stahl, J. Moehl, D. Redmon, J. Sanyal, and B. Bhaduri, "A dataset of recorded electricity outages by united states county 2014–2022," *Scientific Data*, vol. 11, no. 1, p. 271, 2024.